Applied Stochastic Process

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Introduction

A stochastic process is a collection of random variables indexed by time,

$$\{X_t : t \in T\},$$

used to describe the evolution of a random system. Unlike a single random variable, which describes uncertainty at one fixed stage, a stochastic process describes how uncertainty develops dynamically. This distinction makes stochastic processes a natural mathematical language for population growth, queueing systems, reliability models, financial markets, particle motion, and many other time-dependent random phenomena.

The main theme of this course is to understand how probabilistic structure interacts with temporal evolution. In particular, several important classes of stochastic processes are built around a form of "memorylessness." For a Markov process, the future depends on the past only through the present state. For the exponential distribution, the remaining waiting time does not depend on how long one has already waited. These two ideas are closely connected and lead naturally from discrete-time Markov chains to Poisson processes and continuous-time Markov chains.

The first part of the course studies discrete-time Markov chains. Given a state space S, the evolution of the chain is encoded by transition probabilities

$$p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

Repeated transitions are described by powers of the transition matrix. However, the main objective is not merely to compute these probabilities, but to understand the long-term behavior of the chain. This leads to the classification of states by communication, recurrence, transience, periodicity, and positive recurrence. Stationary distributions then provide a probabilistic description of equilibrium and long-run occupation frequencies.

Branching processes provide an important application of discrete-time stochastic processes. They describe populations in which individuals reproduce independently according to a common offspring distribution. Probability generating functions allow the entire evolution of the population to be encoded through function composition. In particular, the extinction problem becomes a fixed-point problem for the generating function, illustrating how analytic methods can reveal qualitative probabilistic behavior.

The exponential distribution and the Poisson process form the next major theme. The memoryless property of the exponential distribution makes it particularly suitable for modeling waiting times between random events. If independent exponential waiting times are accumulated, the resulting counting process is a Poisson process. Its independent and stationary increments, together with superposition and thinning properties, make it one of the fundamental models for randomly occurring events.

Continuous-time Markov chains combine the Markov property with random holding times. The process remains in a state for an exponentially distributed amount of time and then jumps to another state. Transition rates are collected into the generator matrix

$$Q = (q_{ij}),$$

which plays a role analogous to the transition matrix of a discrete-time chain. The Kolmogorov equations describe the evolution of transition probabilities, while stationary and limiting distributions describe long-run behavior. Birth–death processes arise as an especially important class of continuous-time Markov chains.

The final part of the course is devoted to Brownian motion. Brownian motion is a continuous-time stochastic process with continuous sample paths, stationary independent increments, and Gaussian distributions. It can be viewed as the continuous scaling limit of a simple random walk, making it a fundamental bridge between discrete and continuous probability. Its covariance structure,

$$\mathbb{E}[B_s B_t] = s \wedge t,$$

completely determines its finite-dimensional distributions.

Brownian paths exhibit several striking properties. They are continuous but highly irregular: with probability one they are nowhere differentiable and have infinite total variation, while their quadratic variation over $[0, t]$ equals t. These properties sharply distinguish Brownian motion from ordinary smooth functions and explain why new forms of stochastic calculus are required in more advanced study.

Martingales and stopping times provide tools for studying Brownian motion at random times. Hitting times, the reflection principle, running maxima, and zero sets reveal how geometric questions about sample paths can often be translated into precise probabilistic identities. Brownian motion with drift and the Brownian bridge then illustrate how new stochastic processes can be constructed from standard Brownian motion by deterministic transformations or conditioning.

Throughout these notes, emphasis is placed on the principal definitions, theorems, probabilistic interpretations, and structural connections among different stochastic processes. Proofs are included when they are concise and illuminate the underlying mechanism; when a complete proof would require substantial additional machinery, the main idea and mathematical conclusion are emphasized instead. The goal is not only to collect formulas, but to develop a coherent picture of how a small number of fundamental probabilistic principles generate a wide variety of stochastic models.

Chapter 1

Discrete-Time Markov Chains

1 Discrete-Time Markov Chains

1.1 Markov Chains and Transition Probabilities

A discrete-time stochastic process is a collection of random variables

$$\{X_n : n = 0, 1, 2, \ldots\}$$

defined on a common probability space. The set of possible values of X_n is called the *state space* and will usually be denoted by S.

The fundamental idea of a Markov chain is that, once the present state is known, the past contains no additional information about the future.

Definition 1.1 (Markov Property)**.** A stochastic process $\{X_n : n \geq 0\}$ with state space S is called a *discrete-time Markov chain* if, for every $n \geq 0$ and every $i_0, \ldots, i_{n+1} \in S$,

$$\mathbb{P}\left(X_{n+1} = i_{n+1} \mid X_0 = i_0, \ldots, X_n = i_n\right) = \mathbb{P}\left(X_{n+1} = i_{n+1} \mid X_n = i_n\right),$$

whenever the conditional probabilities are well defined.

Thus, conditional on the present state X_n, the distribution of the next state does not depend on the entire previous history.

Throughout this chapter, we mainly consider *time-homogeneous* Markov chains.

Definition 1.2 (Transition Probability)**.** For $i, j \in S$, define

$$p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

If p_{ij} does not depend on n, then the chain is called *time-homogeneous*.

The matrix

$$P = (p_{ij})_{i,j \in S}$$

is called the *transition matrix*. It satisfies

$$p_{ij} \geq 0, \qquad \sum_{j \in S} p_{ij} = 1$$

for every $i \in S$.

Remark 1.3. The transition matrix contains the complete one-step probabilistic description of a time-homogeneous Markov chain. Once P and the distribution of X_0 are specified, the distribution of the entire chain is determined.

1.2 n-Step Transition Probabilities

Definition 1.4 (n-Step Transition Probability)**.** For $n \geq 0$, define

$$p_{ij}^{(n)} = \mathbb{P}(X_n = j \mid X_0 = i).$$

Clearly,

$$p_{ij}^{(0)} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

The n-step transition probabilities are encoded by powers of the transition matrix.

Theorem 1.5. *For every $n \geq 1$,*

$$P^n = (p_{ij}^{(n)})_{i,j \in S}.$$

Proof. The result follows from the Chapman–Kolmogorov equations. For $m, n \geq 0$,

$$p_{ij}^{(m+n)} = \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)}.$$

Indeed, conditioning on the state at time m,

$$\begin{aligned} p_{ij}^{(m+n)} &= \mathbb{P}(X_{m+n} = j \mid X_0 = i) \\ &= \sum_{k \in S} \mathbb{P}(X_{m+n} = j, X_m = k \mid X_0 = i) \\ &= \sum_{k \in S} \mathbb{P}(X_m = k \mid X_0 = i) \mathbb{P}(X_{m+n} = j \mid X_m = k) \\ &= \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)}, \end{aligned}$$

where the last equality follows from time homogeneity and the Markov property. Therefore matrix multiplication gives

$$P^{m+n} = P^m P^n,$$

and hence P^n contains precisely the n-step transition probabilities. □

1.3 Initial Distributions

Suppose the initial distribution of the chain is

$$\alpha = (\alpha_i)_{i \in S}, \qquad \alpha_i = \mathbb{P}(X_0 = i).$$

Then

$$\mathbb{P}(X_n = j) = \sum_{i \in S} \alpha_i p_{ij}^{(n)}.$$

Thus, regarding α as a row vector, the distribution of X_n is

$$\boxed{\alpha P^n}.$$

Example 1.6. Suppose

$$S = \{0, 1\}, \qquad P = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix},$$

and

$$\alpha = (1, 0).$$

Then

$$\mathbb{P}(X_n = j) = (\alpha P^n)_j.$$

Hence the distribution after n steps is obtained simply by multiplying the initial distribution by P^n.

1.4 Accessibility and Communication

To understand the long-run behavior of a Markov chain, it is useful first to determine which states can be reached from which other states.

Definition 1.7 (Accessibility)**.** A state j is said to be *accessible* from i, written

$$i \to j,$$

if there exists some $n \geq 0$ such that

$$p_{ij}^{(n)} > 0.$$

Definition 1.8 (Communication)**.** States i and j are said to *communicate*, written

$$i \leftrightarrow j,$$

if

$$i \to j \qquad \text{and} \qquad j \to i.$$

Communication is an equivalence relation and therefore partitions the state space into disjoint equivalence classes.

Definition 1.9 (Communicating Class)**.** A subset $C \subseteq S$ is called a *communicating class* if every pair of states $i, j \in C$ communicates.

Definition 1.10 (Irreducibility)**.** A Markov chain is called *irreducible* if

$$i \leftrightarrow j \qquad \text{for all } i, j \in S.$$

Equivalently, the entire state space forms a single communicating class.

Example 1.11. Consider

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix}.$$

Every state can eventually reach every other state, so the chain is irreducible.

1.5 Periodicity

Even when a state is recurrent, the times at which the chain can return to that state may possess a rigid arithmetic structure.

Definition 1.12 (Period)**.** For $i \in S$, the *period* of i is

$$d_i = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\}.$$

If $d_i = 1$, then i is called *aperiodic*. Otherwise it is called *periodic*.

Example 1.13. Consider the deterministic two-state chain

$$0 \longleftrightarrow 1.$$

Starting from either state, a return is possible only after $2, 4, 6, \ldots$ steps. Therefore

$$d_0 = d_1 = 2.$$

Periodicity is a class property.

Theorem 1.14. *If*

$$i \leftrightarrow j,$$

then

$$d_i = d_j.$$

Proof. Since $i \leftrightarrow j$, there exist $m, n \geq 0$ such that

$$p_{ij}^{(m)} > 0, \qquad p_{ji}^{(n)} > 0.$$

Suppose

$$p_{jj}^{(r)} > 0.$$

Then the path

$$i \longrightarrow j \longrightarrow j \longrightarrow i$$

has positive probability, so

$$p_{ii}^{(m+r+n)} > 0.$$

Also,

$$p_{ii}^{(m+n)} > 0.$$

By definition, d_i divides every possible return time to i. Hence

$$d_i \mid (m + r + n) \qquad \text{and} \qquad d_i \mid (m + n),$$

which implies

$$d_i \mid r.$$

Since this holds for every r satisfying $p_{jj}^{(r)} > 0$,

$$d_i \mid d_j.$$

By symmetry,

$$d_j \mid d_i.$$

Therefore

$$d_i = d_j.$$

□

Corollary 1.15. *In an irreducible Markov chain, all states have the same period. Therefore it makes sense to call an irreducible chain itself periodic or aperiodic.*

1.6 Recurrence and Transience

The next question is whether the chain eventually returns to a state after leaving it.

For $i \in S$, define the first return time

$$T_i = \inf\{n \geq 1 : X_n = i\}.$$

Definition 1.16 (Recurrence and Transience)**.** A state i is called *recurrent* if

$$\mathbb{P}_i(T_i < \infty) = 1.$$

It is called *transient* if

$$\mathbb{P}_i(T_i < \infty) < 1.$$

Here

$$\mathbb{P}_i(\,\cdot\,) = \mathbb{P}(\,\cdot \mid X_0 = i).$$

Let

$$f_i = \mathbb{P}_i(T_i < \infty)$$

denote the probability of ever returning to i. Thus

$$i \text{ recurrent} \iff f_i = 1,$$

while

$$i \text{ transient} \iff f_i < 1.$$

Another useful viewpoint is to count the total number of visits to a state. Define

$$M_i = \sum_{n=0}^{\infty} \mathbf{1}_{\{X_n = i\}}.$$

Starting from i, the number of returns has a geometric structure.

Proposition 1.17. *For $k \geq 1$,*

$$\mathbb{P}_i(M_i = k) = f_i^{k-1}(1 - f_i).$$

Consequently,

$$\mathbb{P}_i(M_i = \infty) = \begin{cases} 1, & i \text{ is recurrent}, \\ 0, & i \text{ is transient}. \end{cases}$$

Proof. After every visit to i, the Markov property implies that the probability of another future return is again f_i. Hence having exactly k visits means returning $k-1$ times and then never returning again. Therefore

$$\mathbb{P}_i(M_i = k) = f_i^{k-1}(1 - f_i).$$

If $f_i = 1$, the chain returns infinitely many times almost surely. If $f_i < 1$, then

$$\lim_{k \to \infty} f_i^k = 0,$$

so the number of visits is finite almost surely. □

1.7 A Criterion for Recurrence

The return probabilities $p_{ii}^{(n)}$ provide a useful analytic criterion for recurrence.

Theorem 1.18 (Recurrence Criterion)**.** *A state i is recurrent if and only if*

$$\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty.$$

It is transient if and only if

$$\sum_{n=0}^{\infty} p_{ii}^{(n)} < \infty.$$

Proof. Since

$$M_i = \sum_{n=0}^{\infty} \mathbf{1}_{\{X_n = i\}},$$

Tonelli's theorem gives

$$\begin{aligned}\mathbb{E}_i[M_i] &= \sum_{n=0}^{\infty} \mathbb{P}_i(X_n = i) \\ &= \sum_{n=0}^{\infty} p_{ii}^{(n)}.\end{aligned}$$

On the other hand, if $f_i < 1$, then from the geometric distribution above,

$$\mathbb{E}_i[M_i] = \frac{1}{1 - f_i} < \infty.$$

If $f_i = 1$, then $M_i = \infty$ almost surely, so

$$\mathbb{E}_i[M_i] = \infty.$$

Therefore

$$i \text{ recurrent} \iff \sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty.$$

□

1.8 Recurrence as a Class Property

Like periodicity, recurrence and transience are properties shared by communicating states.

Theorem 1.19. *Suppose*

$$i \leftrightarrow j.$$

Then i and j are either both recurrent or both transient.

Proof. Since $i \leftrightarrow j$, there exist $m, r \geq 0$ such that

$$p_{ij}^{(m)} > 0, \qquad p_{ji}^{(r)} > 0.$$

For every $n \geq 0$, the Chapman–Kolmogorov equations give

$$p_{ii}^{(m+n+r)} \geq p_{ij}^{(m)} p_{jj}^{(n)} p_{ji}^{(r)}.$$

Therefore, if

$$\sum_{n=0}^{\infty} p_{jj}^{(n)} = \infty,$$

then

$$\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty.$$

Hence recurrence of j implies recurrence of i.

Interchanging i and j gives the converse. Thus recurrence is a class property, and consequently so is transience. □

Corollary 1.20. *Every finite irreducible Markov chain is recurrent.*

Proof. Suppose, to the contrary, that one state is transient. By irreducibility, every state is then transient. Hence for every $i, j \in S$,

$$p_{ij}^{(n)} \longrightarrow 0 \qquad \text{as } n \to \infty.$$

Since S is finite,

$$1 = \sum_{j \in S} p_{ij}^{(n)} \longrightarrow 0,$$

which is impossible. Therefore every state must be recurrent. □

1.9 Positive and Null Recurrence

Recurrence guarantees eventual return, but it does not guarantee that the expected return time is finite.

Definition 1.21 (Mean Recurrence Time)**.** For a recurrent state i, define

$$m_i = \mathbb{E}_i[T_i].$$

Definition 1.22 (Positive and Null Recurrence)**.** A recurrent state i is called

- *positive recurrent* if
$$m_i < \infty;$$
- *null recurrent* if
$$m_i = \infty.$$

If i is transient, then

$$\mathbb{P}_i(T_i = \infty) > 0,$$

and consequently

$$\mathbb{E}_i[T_i] = \infty.$$

Definition 1.23 (Ergodic State)**.** A state is called *ergodic* if it is both positive recurrent and aperiodic.

For an irreducible chain, positive recurrence is also a class property.

Theorem 1.24. *If i is positive recurrent and*

$$i \leftrightarrow j,$$

then j is positive recurrent.

Remark 1.25. For a finite irreducible Markov chain, every state is positive recurrent. Thus, in the finite irreducible setting, aperiodicity is the only additional condition needed for ergodicity.

1.10 Long-Run Proportion of Visits

For $j \in S$, define the number of visits to j up to time n by

$$N_j(n) = \sum_{k=0}^{n-1} \mathbf{1}_{\{X_k=j\}}.$$

The ratio

$$\frac{N_j(n)}{n}$$

is the proportion of time spent in state j during the first n steps.

For an irreducible recurrent Markov chain, the asymptotic frequency of visits is determined by the mean recurrence time.

Theorem 1.26. *Let the Markov chain be irreducible and recurrent. Then, for every $j \in S$,*

$$\frac{N_j(n)}{n} \longrightarrow \frac{1}{m_j} \qquad \textit{almost surely,}$$

where the convention $1/\infty = 0$ is used.

Remark 1.27. This theorem explains the distinction between positive recurrence and null recurrence. If j is positive recurrent, then

$$\frac{1}{m_j} > 0,$$

so the chain spends a positive fraction of its time in state j. If j is null recurrent, then

$$\frac{1}{m_j} = 0,$$

even though the chain returns to j infinitely often.

1.11 Stationary Distributions

Definition 1.28 (Stationary Distribution)**.** A probability vector

$$\pi = (\pi_j)_{j \in S}$$

is called a *stationary distribution* for the Markov chain if

$$\pi_j \geq 0, \qquad \sum_{j \in S} \pi_j = 1,$$

and

$$\boxed{\pi = \pi P}.$$

Equivalently,

$$\pi_j = \sum_{i \in S} \pi_i p_{ij} \qquad \text{for every } j \in S.$$

The terminology "stationary" comes from the following observation.

Proposition 1.29. *If*

$$X_0 \sim \pi$$

and π is stationary, then

$$X_n \sim \pi \qquad \textit{for every } n \geq 0.$$

Proof. Since the distribution of X_n is

$$\pi P^n,$$

and

$$\pi P = \pi,$$

we obtain inductively

$$\pi P^n = \pi$$

for every $n \geq 0$. □

Remark 1.30. A stationary distribution should not be confused with a limiting distribution. A stationary distribution is a probability vector satisfying

$$\pi P = \pi.$$

A limiting distribution concerns the convergence

$$p_{ij}^{(n)} \longrightarrow \pi_j \qquad (n \to \infty).$$

The existence of a stationary distribution alone does not guarantee this convergence; periodicity can prevent it.

Example 1.31 (Stationary but Not Convergent)**.** Consider

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The stationary distribution is

$$\pi = \left(\frac{1}{2}, \frac{1}{2}\right).$$

However, if $X_0 = 0$, then

$$X_n = \begin{cases} 0, & n \text{ even}, \\ 1, & n \text{ odd}, \end{cases}$$

with probability one. Hence $p_{00}^{(n)}$ oscillates between 1 and 0, so the transition probabilities do not converge. Thus

$$\text{existence of a stationary distribution} \not\Rightarrow \text{convergence to that distribution.}$$

1.12 Stationarity and Positive Recurrence

The existence of a stationary distribution is closely related to positive recurrence.

Theorem 1.32. *For an irreducible Markov chain, the following are equivalent:*

$$\boxed{\textit{the chain has a stationary distribution}}$$

and

$$\boxed{\textit{every state is positive recurrent.}}$$

When these conditions hold, the stationary distribution is unique and is given by

$$\boxed{\pi_j = \frac{1}{m_j}.}$$

Remark 1.33. The formula

$$\pi_j = \frac{1}{m_j}$$

has a natural probabilistic interpretation. The mean time between successive visits to j is m_j, so the long-run frequency of visits to j is its reciprocal.

Corollary 1.34. *Every finite irreducible Markov chain has a unique stationary distribution.*

1.13 Limiting Probabilities

Stationarity describes equilibrium, while limiting probabilities describe whether the chain approaches equilibrium from an arbitrary initial state.

Theorem 1.35 (Limiting Distribution Theorem)**.** *Let $\{X_n\}$ be an irreducible, positive recurrent, and aperiodic Markov chain with stationary distribution π. Then*

$$\boxed{\lim_{n\to\infty} p_{ij}^{(n)} = \pi_j}$$

for every $i, j \in S$.

Thus, in the ergodic case, the limiting distribution does not depend on the initial state.

Remark 1.36. The three assumptions have different roles:

- irreducibility ensures that the state space forms one communicating system;
- positive recurrence guarantees the existence of a stationary probability distribution;
- aperiodicity removes persistent oscillations and allows $p_{ij}^{(n)}$ to converge.

For an irreducible positive recurrent chain, the stationary probability also has the long-run interpretation

$$\pi_j = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \mathbf{1}_{\{X_k=j\}} \qquad \text{almost surely.}$$

Thus three quantities are closely related:

$$\boxed{\pi_j = \frac{1}{m_j} = \text{long-run proportion of time spent in state } j.}$$

If the chain is also aperiodic, then

$$\boxed{\pi_j = \lim_{n\to\infty} p_{ij}^{(n)}.}$$

1.14 Time Reversal

Suppose $\{X_n : 0 \le n \le N\}$ is a Markov chain with transition matrix P, and suppose the chain starts in its stationary distribution π.

Define the reversed process

$$Y_n = X_{N-n}, \qquad 0 \le n \le N.$$

Theorem 1.37 (Time Reversal)**.** *The process $\{Y_n\}$ is also a Markov chain with stationary distribution π. Its transition probabilities are*

$$\boxed{q_{ij} = \frac{\pi_j p_{ji}}{\pi_i}}$$

whenever $\pi_i > 0$.

Proof. By stationarity,

$$\mathbb{P}(X_n = i) = \pi_i$$

for every n. Therefore

$$\begin{aligned} q_{ij} &= \mathbb{P}(Y_{n+1} = j \mid Y_n = i) \\ &= \mathbb{P}(X_{N-n-1} = j \mid X_{N-n} = i) \\ &= \frac{\mathbb{P}(X_{N-n-1} = j, X_{N-n} = i)}{\mathbb{P}(X_{N-n} = i)} \\ &= \frac{\pi_j p_{ji}}{\pi_i}. \end{aligned}$$

Hence the reversed process is again time-homogeneous and Markovian. □

1.15 Reversible Markov Chains

A particularly important situation occurs when the reversed chain has the same transition probabilities as the original chain.

Definition 1.38 (Reversibility)**.** A Markov chain with transition matrix P is called *reversible* with respect to a probability distribution π if

$$\boxed{\pi_i p_{ij} = \pi_j p_{ji}}$$

for all $i, j \in S$. These equations are called the *detailed balance equations.*

Detailed balance has a useful interpretation: in equilibrium, the probability flow from i to j equals the probability flow from j to i.

Theorem 1.39. *If a probability distribution π satisfies the detailed balance equations*

$$\pi_i p_{ij} = \pi_j p_{ji} \qquad \text{for all } i, j \in S,$$

then π is a stationary distribution.

Proof. For every $j \in S$,

$$\begin{aligned} \sum_{i \in S} \pi_i p_{ij} &= \sum_{i \in S} \pi_j p_{ji} \\ &= \pi_j \sum_{i \in S} p_{ji} \\ &= \pi_j. \end{aligned}$$

Therefore

$$\pi P = \pi.$$

□

Remark 1.40. The converse is not true in general:

$$\text{detailed balance} \implies \text{stationarity},$$

but

$$\text{stationarity} \not\implies \text{detailed balance}.$$

Thus reversibility is a stronger property than stationarity.

Example 1.41 (Stationary but Not Reversible)**.** Consider the deterministic cycle

$$1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 1.$$

Its transition matrix is

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

The uniform distribution

$$\pi = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

is stationary.

However,

$$\pi_1 p_{12} = \frac{1}{3}, \qquad \pi_2 p_{21} = 0.$$

Hence detailed balance fails, so the chain is not reversible.

1.16 Summary of Long-Run Behavior

The main concepts introduced in this chapter can be organized as follows.

For states in the same communicating class,

periodicity, recurrence, transience, and positive recurrence are class properties.

For an irreducible chain,

$$\text{stationary distribution exists} \iff \text{positive recurrence.}$$

When the chain is irreducible and positive recurrent,

$$\pi_j = \frac{1}{m_j} = \text{long-run proportion of time spent in state } j.$$

If the chain is additionally aperiodic, then

$$p_{ij}^{(n)} \longrightarrow \pi_j.$$

Finally,

$$\text{detailed balance} \Longrightarrow \text{stationarity},$$

while the reverse implication need not hold.

These results form the basic long-run theory of discrete-time Markov chains and will serve as a foundation for the continuous-time models developed later.

Chapter 2

Branching Processes and Generating Functions

2 Branching Processes and Generating Functions

2.1 Branching Processes

Branching processes provide a simple probabilistic model for the evolution of a population. Each individual independently produces a random number of offspring according to the same probability distribution.

Let

$$\xi$$

be a nonnegative integer-valued random variable with

$$\mathbb{P}(\xi = k) = p_k, \qquad k = 0, 1, 2, \ldots,$$

where

$$p_k \geq 0, \qquad \sum_{k=0}^{\infty} p_k = 1.$$

The random variable ξ represents the number of offspring produced by one individual.

Definition 2.1 (Galton–Watson Branching Process)**.** Let

$$Z_0 = 1,$$

and let

$$\{\xi_{n,k} : n \geq 0,\ k \geq 1\}$$

be independent and identically distributed random variables, each having the same distribution as ξ.

Define recursively

$$\boxed{Z_{n+1} = \sum_{k=1}^{Z_n} \xi_{n,k}.}$$

Then

$$\{Z_n : n \geq 0\}$$

is called a *Galton–Watson branching process.*

Here Z_n denotes the size of the n-th generation.

Conditional on

$$Z_n = m,$$

the next generation consists of the offspring produced independently by the m individuals currently alive:

$$Z_{n+1} = \xi_{n,1} + \cdots + \xi_{n,m}.$$

Remark 2.2. The state 0 is absorbing. Indeed, if

$$Z_n = 0,$$

then there are no individuals capable of producing offspring, and hence

$$Z_{n+1} = 0.$$

Therefore,

$$0 \longrightarrow 0$$

with probability one.

Remark 2.3. The branching process is also a Markov chain on

$$S = \{0, 1, 2, \ldots\}.$$

Given Z_n, the distribution of Z_{n+1} depends only on the current population size and not on the previous generations.

2.2 Probability Generating Functions

Generating functions are particularly effective for studying nonnegative integer-valued random variables.

Definition 2.4 (Probability Generating Function)**.** Let X be a random variable taking values in

$$\{0, 1, 2, \ldots\}.$$

The *probability generating function*, or PGF, of X is

$$\boxed{G_X(s) = \mathbb{E}[s^X] = \sum_{k=0}^{\infty} \mathbb{P}(X = k) s^k,}$$

for values of s for which the series converges.

In particular, for $0 \leq s \leq 1$, the generating function is always well defined. The following elementary properties will be used repeatedly.

Proposition 2.5 (Basic Properties of Generating Functions)**.** *Let X be a nonnegative integer-valued random variable with generating function G_X. Then*

$$G_X(1) = 1,$$

and

$$G_X(0) = \mathbb{P}(X = 0).$$

If the relevant moments are finite, then

$$G'_X(1) = \mathbb{E}[X]$$

and

$$G''_X(1) = \mathbb{E}[X(X-1)].$$

Consequently,

$$\boxed{\mathrm{Var}(X) = G''_X(1) + G'_X(1) - \Big(G'_X(1)\Big)^2.}$$

Proof. From the definition,

$$G_X(1) = \sum_{k=0}^{\infty} \mathbb{P}(X = k) = 1,$$

while

$$G_X(0) = \mathbb{P}(X = 0).$$

Differentiating term by term gives

$$G'_X(s) = \sum_{k=1}^{\infty} k\, \mathbb{P}(X = k) s^{k-1}.$$

Hence

$$G'_X(1) = \sum_{k=1}^{\infty} k\,\mathbb{P}(X = k) = \mathbb{E}[X].$$

Similarly,

$$G''_X(1) = \sum_{k=2}^{\infty} k(k-1)\mathbb{P}(X = k) = \mathbb{E}[X(X-1)].$$

Since

$$X^2 = X(X-1) + X,$$

we have

$$\mathbb{E}[X^2] = G''_X(1) + G'_X(1),$$

and therefore

$$\mathrm{Var}(X) = G''_X(1) + G'_X(1) - \Big(G'_X(1)\Big)^2.$$

□

2.3 Examples of Generating Functions

Example 2.6 (Bernoulli Distribution)**.** If

$$X \sim \text{Bernoulli}(p),$$

then

$$\mathbb{P}(X = 0) = 1 - p, \qquad \mathbb{P}(X = 1) = p.$$

Therefore

$$\boxed{G_X(s) = 1 - p + ps.}$$

Example 2.7 (Poisson Distribution)**.** If

$$X \sim \text{Poisson}(\lambda),$$

then

$$\mathbb{P}(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

Hence

$$\begin{aligned} G_X(s) &= \sum_{k=0}^{\infty} e^{-\lambda}\frac{\lambda^k}{k!}s^k \\ &= e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda s)^k}{k!} \\ &= e^{-\lambda}e^{\lambda s}. \end{aligned}$$

Thus

$$\boxed{G_X(s) = e^{\lambda(s-1)}.}$$

Example 2.8 (Geometric Distribution)**.** Suppose

$$\mathbb{P}(X = k) = p(1-p)^k, \qquad k = 0, 1, 2, \ldots.$$

Then

$$\begin{aligned} G_X(s) &= \sum_{k=0}^{\infty} p(1-p)^k s^k \\ &= p\sum_{k=0}^{\infty}\Big((1-p)s\Big)^k. \end{aligned}$$

Therefore

$$\boxed{G_X(s) = \frac{p}{1-(1-p)s}.}$$

2.4 Generating Functions of Independent Sums

Generating functions transform sums of independent random variables into products.

Proposition 2.9. *If X and Y are independent nonnegative integer-valued random variables, then*

$$\boxed{G_{X+Y}(s) = G_X(s)G_Y(s).}$$

Proof. By independence,

$$\begin{aligned} G_{X+Y}(s) &= \mathbb{E}[s^{X+Y}] \\ &= \mathbb{E}[s^X s^Y] \\ &= \mathbb{E}[s^X]\mathbb{E}[s^Y] \\ &= G_X(s)G_Y(s). \end{aligned}$$

□

More generally, if

$$X_1, \ldots, X_n$$

are independent and identically distributed, then

$$G_{X_1+\cdots+X_n}(s) = \Big(G_X(s)\Big)^n.$$

2.5 Random Sums

A fundamental property of generating functions concerns a random number of independent summands.

Theorem 2.10 (Generating Function of a Random Sum)**.** *Let*

$$X_1, X_2, \ldots$$

be i.i.d. nonnegative integer-valued random variables with generating function G_X. Let N be another nonnegative integer-valued random variable, independent of the X_i, with generating function G_N. Define

$$Y = X_1 + \cdots + X_N,$$

with the convention that $Y = 0$ when $N = 0$. Then

$$\boxed{G_Y(s) = G_N(G_X(s)).}$$

Proof. Conditioning on N,

$$\begin{aligned} G_Y(s) &= \mathbb{E}[s^Y] \\ &= \sum_{n=0}^{\infty} \mathbb{E}[s^Y \mid N = n]\mathbb{P}(N = n). \end{aligned}$$

Conditional on $N = n$,

$$Y = X_1 + \cdots + X_n.$$

By independence,

$$\mathbb{E}[s^Y \mid N = n] = \Big(G_X(s)\Big)^n.$$

Therefore

$$\begin{aligned} G_Y(s) &= \sum_{n=0}^{\infty} \Big(G_X(s)\Big)^n \mathbb{P}(N = n) \\ &= G_N(G_X(s)). \end{aligned}$$

□

Remark 2.11. This composition property is the main reason generating functions are so effective for branching processes: the population of one generation is a random sum of independent offspring counts.

2.6 Generating Functions of a Branching Process

Let

$$G(s) = G_\xi(s) = \mathbb{E}[s^\xi]$$

be the generating function of the offspring distribution.

Let

$$G_n(s) = \mathbb{E}[s^{Z_n}]$$

denote the generating function of the population size in generation n.

Since

$$Z_{n+1} = \sum_{k=1}^{Z_n} \xi_{n,k},$$

the random-sum formula gives

Proposition 2.12. *For every $n \geq 0$,*

$$\boxed{G_{n+1}(s) = G_n(G(s)).}$$

Since $Z_0 = 1$,

$$G_0(s) = s.$$

Therefore

$$G_1(s) = G(s),$$

$$G_2(s) = G(G(s)),$$

and, in general,

$$\boxed{G_n(s) = \underbrace{G \circ G \circ \cdots \circ G}_{n \text{ times}}(s).}$$

Thus the evolution of the branching process is encoded by repeated composition of a single generating function.

2.7 Mean Population Size

Let

$$\mu = \mathbb{E}[\xi] = G'(1).$$

Theorem 2.13. *For a Galton–Watson branching process with $Z_0 = 1$,*

$$\boxed{\mathbb{E}[Z_n] = \mu^n.}$$

Proof. Since

$$G_n(s) = G_{n-1}(G(s)),$$

the chain rule gives

$$G_n'(s) = G_{n-1}'(G(s))G'(s).$$

Evaluating at $s = 1$, and using

$$G(1) = 1, \qquad G'(1) = \mu,$$

we obtain

$$G_n'(1) = \mu G_{n-1}'(1).$$

Since

$$G_n'(1) = \mathbb{E}[Z_n],$$

it follows that

$$\mathbb{E}[Z_n] = \mu \mathbb{E}[Z_{n-1}].$$

Because

$$Z_0 = 1, \qquad \mathbb{E}[Z_0] = 1,$$

iteration gives

$$\boxed{\mathbb{E}[Z_n] = \mu^n.}$$

□

Remark 2.14. The value of μ gives a first indication of the qualitative behavior of the population:

$$\mu < 1$$

corresponds to an average decrease,

$$\mu = 1$$

to an average critical balance, and

$$\mu > 1$$

to an average increase.

2.8 Variance of the Population Size

Let

$$\sigma^2 = \mathrm{Var}(\xi).$$

Theorem 2.15. *For a Galton–Watson branching process with $Z_0 = 1$,*

$$\boxed{\mathrm{Var}(Z_n) = \begin{cases} n\sigma^2, & \mu = 1, \\ \sigma^2 \mu^{n-1} \dfrac{\mu^n - 1}{\mu - 1}, & \mu \neq 1. \end{cases}}$$

Proof. Since

$$G_n(s) = G_{n-1}(G(s)),$$

differentiating twice yields

$$G_n''(s) = G_{n-1}''(G(s))\Big(G'(s)\Big)^2 + G_{n-1}'(G(s))G''(s).$$

Evaluating at $s = 1$,

$$G_n''(1) = \mu^2 G_{n-1}''(1) + \mu^{n-1} G''(1),$$

because

$$G_{n-1}'(1) = \mathbb{E}[Z_{n-1}] = \mu^{n-1}.$$

Also,

$$G''(1) = \mathbb{E}[\xi(\xi - 1)].$$

Since

$$\sigma^2 = \mathrm{Var}(\xi) = G''(1) + \mu - \mu^2,$$

we have

$$G''(1) = \sigma^2 - \mu + \mu^2.$$

Using

$$\mathrm{Var}(Z_n) = G_n''(1) + \mathbb{E}[Z_n] - \Big(\mathbb{E}[Z_n]\Big)^2,$$

together with

$$\mathbb{E}[Z_n] = \mu^n,$$

one obtains the recursion

$$\boxed{\mathrm{Var}(Z_n) = \mu^2 \,\mathrm{Var}(Z_{n-1}) + \sigma^2 \mu^{n-1}.}$$

If $\mu = 1$, then

$$\mathrm{Var}(Z_n) = \mathrm{Var}(Z_{n-1}) + \sigma^2.$$

Since

$$\mathrm{Var}(Z_0) = 0,$$

we obtain

$$\mathrm{Var}(Z_n) = n\sigma^2.$$

If $\mu \neq 1$, iterating the recursion gives

$$\begin{aligned} \mathrm{Var}(Z_n) &= \sigma^2 \left(\mu^{2n-2} + \mu^{2n-3} + \cdots + \mu^{n-1}\right) \\ &= \sigma^2 \mu^{n-1} \left(1 + \mu + \cdots + \mu^{n-1}\right). \end{aligned}$$

Therefore

$$\mathrm{Var}(Z_n) = \sigma^2 \mu^{n-1} \frac{\mu^n - 1}{\mu - 1}.$$

□

2.9 Extinction

Because the state 0 is absorbing, once the population becomes extinct, it remains extinct forever.

Definition 2.16 (Extinction Event)**.** The event

$$\mathcal{E} = \{\text{there exists } n \geq 0 \text{ such that } Z_n = 0\}$$

is called the *extinction event*. Its probability

$$\alpha = \mathbb{P}(\mathcal{E} \mid Z_0 = 1)$$

is called the *extinction probability*.

Since 0 is absorbing,

$$\{Z_n = 0\} \subseteq \{Z_{n+1} = 0\}.$$

Hence

$$\mathbb{P}(Z_n = 0)$$

is nondecreasing in n, and

$$\alpha = \lim_{n\to\infty} \mathbb{P}(Z_n = 0).$$

But

$$\mathbb{P}(Z_n = 0) = G_n(0).$$

Therefore

$$\boxed{\alpha = \lim_{n\to\infty} G_n(0).}$$

2.10 The Fixed-Point Equation for Extinction

The extinction probability is characterized by a fixed point of the offspring generating function.

Theorem 2.17 (Extinction Probability)**.** *Let*

$$G(s) = \mathbb{E}[s^\xi]$$

be the generating function of the offspring distribution.

Then the extinction probability α is the smallest solution in $[0, 1]$ of

$$\boxed{s = G(s).}$$

Proof. Define

$$q_n = \mathbb{P}(Z_n = 0) = G_n(0).$$

Since

$$G_{n+1}(s) = G_n(G(s)),$$

we may equivalently write, using repeated composition,

$$q_{n+1} = G(q_n), \qquad q_0 = 0.$$

The sequence $\{q_n\}$ is nondecreasing and bounded above by 1. Therefore there exists

$$\alpha = \lim_{n\to\infty} q_n.$$

Since G is continuous on $[0, 1]$,

$$\alpha = \lim_{n\to\infty} q_{n+1} = \lim_{n\to\infty} G(q_n) = G(\alpha).$$

Thus α is a solution of

$$s = G(s).$$

Now suppose $r \in [0, 1]$ is any other solution:

$$G(r) = r.$$

Since

$$q_0 = 0 \leq r,$$

and G is increasing on $[0,1]$,

$$q_1 = G(q_0) \leq G(r) = r.$$

Inductively,

$$q_n \leq r$$

for every n. Passing to the limit gives

$$\alpha \leq r.$$

Hence α is the smallest solution of

$$s = G(s)$$

in $[0,1]$. □

2.11 Subcritical, Critical, and Supercritical Cases

The mean offspring number

$$\mu = \mathbb{E}[\xi]$$

determines whether extinction occurs almost surely.

Theorem 2.18 (Extinction Classification)**.** *Let*

$$\mu = \mathbb{E}[\xi], \qquad \sigma^2 = \mathrm{Var}(\xi).$$

Then:

1. *If*
$$\mu < 1,$$
then
$$\boxed{\alpha = 1}.$$
2. *If*
$$\mu = 1 \qquad \textit{and} \qquad \sigma^2 > 0,$$
then
$$\boxed{\alpha = 1}.$$
3. *If*
$$\mu > 1,$$
then
$$\boxed{0 \leq \alpha < 1}.$$

Proof. Recall that

$$G(1) = 1$$

and

$$G'(1) = \mu.$$

Furthermore,

$$G''(s) \geq 0$$

for $0 \leq s \leq 1$, so G is convex.

Consider

$$H(s) = G(s) - s.$$

The extinction probabilities are precisely the zeros of H in $[0, 1]$.

First suppose

$$\mu < 1.$$

Then

$$H'(1) = G'(1) - 1 = \mu - 1 < 0.$$

By convexity of G, the graph of G cannot cross the line $y = s$ at a point strictly below 1. Hence the only fixed point in $[0, 1]$ is

$$s = 1,$$

so

$$\alpha = 1.$$

Now suppose

$$\mu = 1, \qquad \sigma^2 > 0.$$

Then the offspring distribution is nondegenerate, so G is strictly convex. Since

$$G(1) = 1 \qquad \text{and} \qquad G'(1) = 1,$$

strict convexity implies

$$G(s) > s \qquad \text{for } 0 \le s < 1.$$

Thus again the only fixed point is

$$s = 1,$$

and therefore

$$\alpha = 1.$$

Finally suppose

$$\mu > 1.$$

Then

$$H'(1) = \mu - 1 > 0.$$

Hence for $s < 1$ sufficiently close to 1,

$$H(s) < 0.$$

On the other hand,

$$H(0) = G(0) = p_0 \ge 0.$$

By continuity, H has a zero in $[0, 1)$. Therefore the smallest fixed point satisfies

$$\alpha < 1.$$

□

2.12 Interpretation of the Three Regimes

The branching process is classified according to the expected number of offspring per individual.

Definition 2.19. A Galton–Watson branching process is called

- *subcritical* if
$$\mu < 1;$$
- *critical* if
$$\mu = 1;$$
- *supercritical* if
$$\mu > 1.$$

In the subcritical case,
$$\mathbb{E}[Z_n] = \mu^n \longrightarrow 0,$$
and extinction occurs with probability one.

In the critical nondegenerate case,
$$\mathbb{E}[Z_n] = 1$$
for every n, but extinction still occurs with probability one.

In the supercritical case,
$$\mathbb{E}[Z_n] = \mu^n \longrightarrow \infty,$$
and there is a strictly positive probability that the population survives forever.

Remark 2.20. The critical case illustrates an important distinction between expectation and typical behavior. Although
$$\mathbb{E}[Z_n] = 1$$
for every generation, this does not imply that the population remains close to one individual. In the nondegenerate critical case, extinction still occurs almost surely.

2.13 Summary

For a Galton–Watson branching process
$$Z_{n+1} = \sum_{k=1}^{Z_n} \xi_{n,k},$$
the probability generating function
$$G(s) = \mathbb{E}[s^{\xi}]$$
encodes the offspring distribution.

The generating function of the n-th generation is obtained by iteration:
$$\boxed{G_n(s) = G^{\circ n}(s).}$$

If
$$\mu = \mathbb{E}[\xi], \qquad \sigma^2 = \mathrm{Var}(\xi),$$
then
$$\boxed{\mathbb{E}[Z_n] = \mu^n,}$$

and

$$\boxed{\mathrm{Var}(Z_n) = \begin{cases} n\sigma^2, & \mu = 1, \\ \sigma^2 \mu^{n-1} \dfrac{\mu^n - 1}{\mu - 1}, & \mu \neq 1. \end{cases}}$$

Finally, the extinction probability α is characterized by

$$\boxed{\alpha = \text{the smallest solution of } s = G(s) \text{ in } [0, 1].}$$

Consequently,

$$\boxed{\begin{array}{rcl} \mu < 1 & \Longrightarrow & \alpha = 1, \\ \mu = 1,\ \sigma^2 > 0 & \Longrightarrow & \alpha = 1, \\ \mu > 1 & \Longrightarrow & \alpha < 1. \end{array}}$$

Chapter 3

Exponential Distributions and Poisson Processes

3 Exponential Distributions and Poisson Processes

3.1 The Exponential Distribution

The exponential distribution is a fundamental continuous distribution for modeling waiting times between random events.

Definition 3.1 (Exponential Distribution)**.** A nonnegative random variable X is said to have an exponential distribution with rate $\lambda > 0$, written

$$X \sim \mathrm{Exp}(\lambda),$$

if its probability density function is

$$f_X(x) = \lambda e^{-\lambda x}, \qquad x \geq 0.$$

Its distribution function is

$$F_X(x) = 1 - e^{-\lambda x}, \qquad x \geq 0,$$

and hence

$$\boxed{\mathbb{P}(X > t) = e^{-\lambda t}, \qquad t \geq 0.}$$

Proposition 3.2. *If*

$$X \sim \mathrm{Exp}(\lambda),$$

then

$$\boxed{\mathbb{E}[X] = \frac{1}{\lambda}, \qquad \mathrm{Var}(X) = \frac{1}{\lambda^2}.}$$

Proof. By definition,

$$\mathbb{E}[X] = \int_0^\infty x\lambda e^{-\lambda x}\,\mathrm{d}x = \frac{1}{\lambda}.$$

Similarly,

$$\mathbb{E}[X^2] = \int_0^\infty x^2\lambda e^{-\lambda x}\,\mathrm{d}x = \frac{2}{\lambda^2}.$$

Hence

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \Big(\mathbb{E}[X]\Big)^2 = \frac{1}{\lambda^2}.$$

□

3.2 The Memoryless Property

The characteristic property of the exponential distribution is its memorylessness.

Theorem 3.3 (Memoryless Property)**.** *If*

$$X \sim \mathrm{Exp}(\lambda),$$

then for every $s, t \geq 0$,

$$\boxed{\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).}$$

Proof. Since

$$\{X > s+t\} \subseteq \{X > s\},$$

we have

$$\begin{aligned}\mathbb{P}(X > s+t \mid X > s) &= \frac{\mathbb{P}(X > s+t)}{\mathbb{P}(X > s)} \\ &= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} \\ &= e^{-\lambda t} \\ &= \mathbb{P}(X > t).\end{aligned}$$

□

Remark 3.4. The memoryless property means that after waiting for s units of time without an event occurring, the distribution of the remaining waiting time is the same as it was at time 0.

3.3 Sums of Exponential Random Variables

Suppose

$$X_1, \ldots, X_n$$

are independent random variables satisfying

$$X_i \sim \text{Exp}(\lambda).$$

Define

$$S_n = X_1 + \cdots + X_n.$$

Theorem 3.5 (Gamma Distribution)**.** *The random variable S_n has density*

$$\boxed{f_{S_n}(t) = \frac{\lambda^n t^{n-1}}{(n-1)!} e^{-\lambda t}, \qquad t \geq 0.}$$

Thus

$$S_n \sim \text{Gamma}(n, \lambda).$$

Remark 3.6. When the shape parameter n is a positive integer, the Gamma distribution is also called the *Erlang distribution.*

Since

$$S_n = X_1 + \cdots + X_n,$$

we immediately have

$$\boxed{\mathbb{E}[S_n] = \frac{n}{\lambda}, \qquad \text{Var}(S_n) = \frac{n}{\lambda^2}.}$$

3.4 Competing Exponential Random Variables

Independent exponential random variables can be interpreted as independent clocks running simultaneously.

Proposition 3.7. *Let*

$$X_1 \sim \text{Exp}(\lambda_1), \qquad X_2 \sim \text{Exp}(\lambda_2),$$

and suppose X_1 and X_2 are independent. Then

$$\boxed{\mathbb{P}(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}.}$$

Proof. Conditioning on $X_1 = x$,

$$\begin{aligned}\mathbb{P}(X_1 < X_2) &= \int_0^\infty \mathbb{P}(X_2 > x) f_{X_1}(x)\,\mathrm{d}x \\ &= \int_0^\infty e^{-\lambda_2 x}\lambda_1 e^{-\lambda_1 x}\,\mathrm{d}x \\ &= \lambda_1 \int_0^\infty e^{-(\lambda_1+\lambda_2)x}\,\mathrm{d}x \\ &= \frac{\lambda_1}{\lambda_1+\lambda_2}.\end{aligned}$$

□

Theorem 3.8 (Minimum of Independent Exponentials)**.** *Let*

$$X_i \sim \mathrm{Exp}(\lambda_i), \qquad i = 1, \ldots, n,$$

be independent, and define

$$X = \min\{X_1, \ldots, X_n\}.$$

Then

$$\boxed{X \sim \mathrm{Exp}\left(\lambda_1 + \cdots + \lambda_n\right).}$$

Proof. For $t \geq 0$,

$$\begin{aligned}\mathbb{P}(X > t) &= \mathbb{P}(X_1 > t, \ldots, X_n > t) \\ &= \prod_{i=1}^n \mathbb{P}(X_i > t) \\ &= \prod_{i=1}^n e^{-\lambda_i t} \\ &= e^{-(\lambda_1+\cdots+\lambda_n)t}.\end{aligned}$$

Hence X has an exponential distribution with rate

$$\lambda_1 + \cdots + \lambda_n.$$

□

Proposition 3.9. *Under the same assumptions,*

$$\boxed{\mathbb{P}\left(X_i = \min_{1\leq j\leq n} X_j\right) = \frac{\lambda_i}{\lambda_1+\cdots+\lambda_n}.}$$

Proof. Using independence,

$$\begin{aligned}\mathbb{P}\left(X_i = \min_j X_j\right) &= \int_0^\infty f_{X_i}(t) \prod_{j\neq i} \mathbb{P}(X_j > t)\,\mathrm{d}t \\ &= \int_0^\infty \lambda_i e^{-\lambda_i t} \prod_{j\neq i} e^{-\lambda_j t}\,\mathrm{d}t \\ &= \lambda_i \int_0^\infty e^{-(\lambda_1+\cdots+\lambda_n)t}\,\mathrm{d}t \\ &= \frac{\lambda_i}{\lambda_1+\cdots+\lambda_n}.\end{aligned}$$

□

Remark 3.10. The minimum result gives the waiting time until the first of several possible events occurs, while

$$\frac{\lambda_i}{\sum_j \lambda_j}$$

gives the probability that event i occurs first. This principle will be used repeatedly in continuous-time Markov chains.

3.5 Construction of the Poisson Process

Suppose

$$T_1, T_2, \ldots \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda), \qquad \lambda > 0.$$

Define the arrival times

$$S_n = T_1 + \cdots + T_n, \qquad n \geq 1,$$

with $S_0 = 0$.

Here T_n is the waiting time between the $(n-1)$-st and n-th events, while S_n is the time of the n-th event. Since S_n is the sum of n independent exponential random variables,

$$S_n \sim \text{Gamma}(n, \lambda),$$

with density

$$\boxed{f_{S_n}(t) = \frac{\lambda^n t^{n-1}}{(n-1)!} e^{-\lambda t}, \qquad t \geq 0.}$$

Definition 3.11 (Poisson Process)**.** For $t \geq 0$, define

$$\boxed{N(t) = \max\{n \geq 0 : S_n \leq t\}.}$$

Then

$$\{N(t) : t \geq 0\}$$

is called a *Poisson process with rate* λ.

The relation between the counting process and its arrival times is

$$\boxed{\{N(t) \geq n\} = \{S_n \leq t\}.}$$

3.6 Infinitesimal Characterization

The Poisson process can also be described directly through its local behavior over short time intervals.

Definition 3.12 (Infinitesimal Characterization)**.** A counting process

$$\{N(t) : t \geq 0\}$$

is a Poisson process with rate $\lambda > 0$ if:

1. $$N(0) = 0;$$
2. it has independent increments;

3. for every $t \geq 0$, as $h \downarrow 0$,

$$\mathbb{P}\Big(N(t+h) - N(t) = 1\Big) = \lambda h + o(h),$$

and

$$\mathbb{P}\Big(N(t+h) - N(t) \geq 2\Big) = o(h).$$

Here

$$\frac{o(h)}{h} \longrightarrow 0 \qquad \text{as } h \downarrow 0.$$

The first short-time condition means that exactly one event occurs during an interval of length h with probability approximately λh. The second condition means that two or more events have probability of smaller order than h.

3.7 Poisson Increments

An equivalent characterization uses the distribution of increments.

Theorem 3.13 (Poisson Increment Characterization)**.** *A counting process $\{N(t) : t \geq 0\}$ is a Poisson process with rate λ if*

1. $$N(0) = 0;$$
2. *it has independent increments;*
3. *for every $0 \leq s < t$,*

$$\boxed{N(t) - N(s) \sim \text{Poisson}\Big(\lambda(t-s)\Big).}$$

Taking $s = 0$,

$$\boxed{N(t) \sim \text{Poisson}(\lambda t).}$$

Hence

$$\mathbb{P}(N(t) = n) = e^{-\lambda t}\frac{(\lambda t)^n}{n!}, \qquad n = 0, 1, 2, \ldots,$$

and therefore

$$\boxed{\mathbb{E}[N(t)] = \text{Var}(N(t)) = \lambda t.}$$

3.8 Independent and Stationary Increments

Definition 3.14 (Independent Increments)**.** A stochastic process $\{N(t) : t \geq 0\}$ has *independent increments* if, for every

$$0 = t_0 < t_1 < \cdots < t_m,$$

the random variables

$$N(t_1) - N(t_0), \quad N(t_2) - N(t_1), \quad \ldots, \quad N(t_m) - N(t_{m-1})$$

are independent.

Definition 3.15 (Stationary Increments)**.** A stochastic process has *stationary increments* if the distribution of

$$N(t+s) - N(s)$$

depends only on t and not on s.

For a Poisson process,

$$N(t+s) - N(s) \sim \text{Poisson}(\lambda t).$$

Thus a Poisson process has both independent and stationary increments.

3.9 Superposition of Poisson Processes

Independent Poisson processes can be combined into another Poisson process.

Theorem 3.16 (Superposition)**.** *Let*

$$\{N_1(t) : t \geq 0\}$$

and

$$\{N_2(t) : t \geq 0\}$$

be independent Poisson processes with rates λ_1 and λ_2. Define

$$N(t) = N_1(t) + N_2(t).$$

Then

$$\boxed{\{N(t) : t \geq 0\} \text{ is a Poisson process with rate } \lambda_1 + \lambda_2.}$$

Proof. For $0 \leq s < t$,

$$N(t) - N(s) = \Big(N_1(t) - N_1(s)\Big) + \Big(N_2(t) - N_2(s)\Big).$$

The two terms are independent and satisfy

$$N_1(t) - N_1(s) \sim \text{Poisson}\Big(\lambda_1(t-s)\Big),$$

and

$$N_2(t) - N_2(s) \sim \text{Poisson}\Big(\lambda_2(t-s)\Big).$$

Hence

$$N(t) - N(s) \sim \text{Poisson}\Big((\lambda_1 + \lambda_2)(t-s)\Big).$$

Independence of increments follows from the independence of the two original processes. Therefore N is a Poisson process with rate

$$\lambda_1 + \lambda_2.$$

□

More generally, the superposition of independent Poisson processes with rates

$$\lambda_1, \ldots, \lambda_k$$

is a Poisson process with rate

$$\lambda_1 + \cdots + \lambda_k.$$

3.10 Thinning of a Poisson Process

Suppose

$$\{N(t) : t \geq 0\}$$

is a Poisson process with rate λ. Independently classify each arrival as Type I with probability p and Type II with probability $1-p$. Let $N_1(t)$ and $N_2(t)$ denote the numbers of Type I and Type II events by time t.

Theorem 3.17 (Thinning Theorem)**.** *The processes*

$$\{N_1(t) : t \geq 0\} \qquad \textit{and} \qquad \{N_2(t) : t \geq 0\}$$

are independent Poisson processes with respective rates

$$\boxed{\lambda p \qquad \textit{and} \qquad \lambda(1-p).}$$

Proof. Conditional on

$$N(t) = n,$$

each of the n events is independently classified as Type I with probability p. Hence

$$N_1(t) \mid N(t) = n \sim \text{Binomial}(n, p).$$

For $k, \ell \geq 0$,

$$\begin{aligned}
&\mathbb{P}\Big(N_1(t) = k, N_2(t) = \ell\Big) \\
&= \mathbb{P}(N(t) = k + \ell)\binom{k+\ell}{k} p^k (1-p)^\ell \\
&= e^{-\lambda t}\frac{(\lambda t)^{k+\ell}}{(k+\ell)!}\frac{(k+\ell)!}{k!\ell!} p^k (1-p)^\ell \\
&= \left[e^{-\lambda p t}\frac{(\lambda p t)^k}{k!}\right]\left[e^{-\lambda(1-p)t}\frac{(\lambda(1-p)t)^\ell}{\ell!}\right].
\end{aligned}$$

Therefore

$$N_1(t) \sim \text{Poisson}(\lambda p t), \qquad N_2(t) \sim \text{Poisson}(\lambda(1-p)t),$$

and the factorization proves their independence. □

3.11 Order Statistics

Suppose

$$Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} \text{Unif}(0, t).$$

Let

$$Y_{(1)} < Y_{(2)} < \cdots < Y_{(n)}$$

denote their order statistics.

Proposition 3.18. *The joint density of*

$$Y_{(1)}, \ldots, Y_{(n)}$$

is

$$\boxed{f_{Y_{(1)},\ldots,Y_{(n)}}(y_1, \ldots, y_n) = \frac{n!}{t^n},}$$

for

$$0 < y_1 < y_2 < \cdots < y_n < t,$$

and is zero otherwise.

3.12 Conditional Distribution of Arrival Times

Let

$$S_1 < S_2 < \cdots$$

be the successive arrival times of a Poisson process with rate λ.

Theorem 3.19 (Conditional Arrival Times)**.** *Given*

$$N(t) = n,$$

the arrival times satisfy

$$\boxed{(S_1, \ldots, S_n) \mid \{N(t) = n\} \overset{\mathrm{d}}{=} (Y_{(1)}, \ldots, Y_{(n)}),}$$

where

$$Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} \mathrm{Unif}(0, t).$$

Proof. For

$$0 < s_1 < \cdots < s_n < t,$$

the probability density corresponding to arrivals near $s_1, \ldots, s_n$ and no further arrival before time t is

$$\lambda^n e^{-\lambda t}.$$

Thus

$$\mathbb{P}\Big(S_1 \in \mathrm{d}s_1, \ldots, S_n \in \mathrm{d}s_n, N(t) = n\Big) = \lambda^n e^{-\lambda t}\, \mathrm{d}s_1 \cdots \mathrm{d}s_n.$$

Since

$$\mathbb{P}(N(t) = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!},$$

the conditional joint density is

$$\frac{\lambda^n e^{-\lambda t}}{e^{-\lambda t}(\lambda t)^n / n!} = \frac{n!}{t^n}.$$

This is exactly the joint density of the order statistics

$$Y_{(1)}, \ldots, Y_{(n)}.$$

Hence

$$(S_1, \ldots, S_n) \mid \{N(t) = n\} \overset{\mathrm{d}}{=} (Y_{(1)}, \ldots, Y_{(n)}).$$

□

Remark 3.20. Conditional on exactly n events occurring by time t, the n arrival times behave as n independent points sampled uniformly from $(0, t)$ and then arranged in increasing order. In particular, the conditional distribution no longer depends on λ.

3.13 Time-Dependent Classification of Events

Suppose

$$\{N(t) : t \geq 0\}$$

is a Poisson process with rate λ. An event occurring at time s is independently classified as Type i, where

$$i = 1, \ldots, K,$$

with probability

$$p_i(s), \qquad \sum_{i=1}^{K} p_i(s) = 1.$$

Let

$$N_i(t)$$

denote the number of Type i events occurring by time t.

Theorem 3.21 (Time-Dependent Classification)**.** *The random variables*

$$N_1(t), \ldots, N_K(t)$$

are independent Poisson random variables satisfying

$$\boxed{N_i(t) \sim \text{Poisson}(\lambda_i(t)),}$$

where

$$\boxed{\lambda_i(t) = \lambda \int_0^t p_i(s)\, \mathrm{d}s.}$$

Remark 3.22. If

$$p_i(s) = p_i$$

does not depend on time, then

$$\lambda_i(t) = \lambda p_i t.$$

Thus ordinary thinning is recovered as a special case.

3.14 Summary

The exponential distribution is characterized by the memoryless property

$$\boxed{\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).}$$

For independent exponential random variables

$$X_i \sim \text{Exp}(\lambda_i),$$

the first event time satisfies

$$\boxed{\min_i X_i \sim \text{Exp}\left(\sum_i \lambda_i\right),}$$

and

$$\boxed{\mathbb{P}\left(X_i = \min_j X_j\right) = \frac{\lambda_i}{\sum_j \lambda_j}.}$$

A Poisson process with rate λ can be constructed from i.i.d. exponential interarrival times:

$$T_i \sim \text{Exp}(\lambda), \qquad S_n = T_1 + \cdots + T_n, \qquad N(t) = \max\{n : S_n \leq t\}.$$

Its increments satisfy

$$\boxed{N(t) - N(s) \sim \text{Poisson}\Big(\lambda(t-s)\Big).}$$

Independent Poisson processes obey the superposition principle

$$\boxed{\lambda_1, \lambda_2 \longrightarrow \lambda_1 + \lambda_2,}$$

while thinning gives

$$\boxed{\lambda \longrightarrow \lambda p \quad \text{and} \quad \lambda(1-p).}$$

Finally, conditional on

$$N(t) = n,$$

the arrival times have the uniform order-statistic representation

$$\boxed{(S_1, \ldots, S_n) \mid \{N(t) = n\} \overset{\text{d}}{=} (Y_{(1)}, \ldots, Y_{(n)}),}$$

where

$$Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} \text{Unif}(0, t).$$

Chapter 4

Continuous-Time Markov Chains

4 Continuous-Time Markov Chains

4.1 Continuous-Time Markov Chains

A continuous-time Markov chain extends the Markov property from discrete time to continuous time. The process may remain in one state for a random amount of time and then jump to another state.

Definition 4.1 (Continuous-Time Markov Chain)**.** A stochastic process

$$\{X(t) : t \geq 0\}$$

with countable state space S is called a *continuous-time Markov chain* if, for every $s, t \geq 0$ and $i, j \in S$,

$$\mathbb{P}\Big(X(t+s) = j \mid X(s) = i, \{X(u) : 0 \leq u < s\}\Big) = \mathbb{P}\Big(X(t+s) = j \mid X(s) = i\Big).$$

For a time-homogeneous chain, the right-hand side depends only on t, and we write

$$\boxed{p_{ij}(t) = \mathbb{P}(X(t) = j \mid X(0) = i).}$$

The matrix

$$P(t) = \Big(p_{ij}(t)\Big)_{i,j \in S}$$

is called the *transition probability matrix* at time t.

For every $t \geq 0$,

$$p_{ij}(t) \geq 0, \qquad \sum_{j \in S} p_{ij}(t) = 1,$$

and at time 0,

$$P(0) = I.$$

4.2 The Chapman–Kolmogorov Equations

The transition probabilities satisfy the continuous-time version of the Chapman–Kolmogorov equations.

Theorem 4.2 (Chapman–Kolmogorov Equations)**.** *For every $s, t \geq 0$,*

$$\boxed{p_{ij}(s+t) = \sum_{k \in S} p_{ik}(s) p_{kj}(t).}$$

Equivalently,

$$\boxed{P(s+t) = P(s)P(t).}$$

Proof. Conditioning on the state at time s,

$$\begin{aligned} p_{ij}(s+t) &= \mathbb{P}(X(s+t) = j \mid X(0) = i) \\ &= \sum_{k \in S} \mathbb{P}\Big(X(s+t) = j, X(s) = k \mid X(0) = i\Big) \\ &= \sum_{k \in S} \mathbb{P}(X(s) = k \mid X(0) = i) \mathbb{P}(X(s+t) = j \mid X(s) = k) \\ &= \sum_{k \in S} p_{ik}(s) p_{kj}(t), \end{aligned}$$

where the last equality follows from the Markov property and time homogeneity. □

Remark 4.3. The family

$$\{P(t) : t \geq 0\}$$

forms a semigroup:

$$P(0) = I, \qquad P(s+t) = P(s)P(t).$$

This is the continuous-time analogue of

$$P^{m+n} = P^m P^n$$

for a discrete-time Markov chain.

4.3 Holding Times

Suppose the process is currently in state i. Let T_i denote the amount of time the process remains in i before making its next jump.

The Markov property implies that the remaining holding time cannot depend on how long the process has already remained in the state. Therefore the holding time must have the memoryless property.

Theorem 4.4 (Exponential Holding Times)**.** *For a continuous-time Markov chain, the holding time in state i is exponentially distributed:*

$$\boxed{T_i \sim \text{Exp}(\nu_i)}$$

for some rate $\nu_i \geq 0$.

Remark 4.5. The parameter ν_i is the total rate of leaving state i. Thus

$$\mathbb{E}[T_i] = \frac{1}{\nu_i}$$

whenever $\nu_i > 0$.

A large value of ν_i means that the process tends to leave state i quickly, while a small value corresponds to a longer expected holding time.

If

$$\nu_i = 0,$$

the process never leaves state i, so i is absorbing.

4.4 The Embedded Jump Chain

A continuous-time Markov chain contains an underlying discrete-time Markov chain that records only the sequence of states visited at jump times.

Let

$$0 = S_0 < S_1 < S_2 < \cdots$$

denote the successive jump times, and define

$$Y_n = X(S_n).$$

Definition 4.6 (Embedded Jump Chain)**.** The process

$$\{Y_n : n \geq 0\}$$

is called the *embedded jump chain* of the continuous-time Markov chain.

If the process is currently in state i, define

$$p_{ij} = \mathbb{P}\left(\text{next state is } j \mid \text{current state is } i\right), \qquad j \neq i.$$

Then

$$\sum_{j \neq i} p_{ij} = 1$$

whenever $\nu_i > 0$.

Thus a continuous-time Markov chain can be described by two pieces of information:

$$\boxed{\begin{array}{c}\text{holding rate } \nu_i, \\ \text{jump probabilities } p_{ij}.\end{array}}$$

The holding rate determines *when* the next jump occurs, while the embedded chain determines *where* the process jumps.

4.5 Transition Rates and the Generator

Instead of specifying holding rates and jump probabilities separately, it is often more convenient to combine them into transition rates.

For $i \neq j$, define

$$\boxed{q_{ij} = \nu_i p_{ij}.}$$

The quantity q_{ij} is the instantaneous transition rate from state i to state j.

Define the diagonal terms by

$$\boxed{q_{ii} = -\sum_{j \neq i} q_{ij} = -\nu_i.}$$

Definition 4.7 (Generator)**.** The matrix

$$\boxed{Q = (q_{ij})_{i,j \in S}}$$

is called the *generator*, or *infinitesimal generator*, of the continuous-time Markov chain.

The generator satisfies

$$q_{ij} \geq 0 \qquad (i \neq j),$$

and

$$\boxed{\sum_{j \in S} q_{ij} = 0.}$$

Conversely, from the generator we recover

$$\nu_i = -q_{ii},$$

and, when $\nu_i > 0$,

$$\boxed{p_{ij} = \frac{q_{ij}}{\nu_i} = \frac{q_{ij}}{-q_{ii}}, \qquad j \neq i.}$$

Thus the descriptions

$$(\nu_i, p_{ij}) \qquad \text{and} \qquad Q$$

contain the same information.

4.6 Infinitesimal Transition Probabilities

The entries of the generator describe the behavior of the process over a very short interval.

Proposition 4.8. *As $h \downarrow 0$, for $i \neq j$,*

$$\boxed{p_{ij}(h) = q_{ij}h + o(h),}$$

while

$$\boxed{p_{ii}(h) = 1 + q_{ii}h + o(h) = 1 - \nu_i h + o(h).}$$

Equivalently,

$$q_{ij} = \lim_{h \downarrow 0} \frac{p_{ij}(h)}{h}, \qquad i \neq j,$$

and

$$q_{ii} = \lim_{h \downarrow 0} \frac{p_{ii}(h) - 1}{h}.$$

In matrix form,

$$\boxed{P(h) = I + hQ + o(h).}$$

Therefore

$$\boxed{Q = \lim_{h \downarrow 0} \frac{P(h) - I}{h}.}$$

4.7 Competing Exponential Clocks

The transition-rate interpretation can also be obtained from competing exponential clocks.

Suppose that while the process is in state i, each possible transition

$$i \longrightarrow j, \qquad j \neq i,$$

is associated with an independent exponential clock of rate q_{ij}.

From the results of the previous chapter,

$$\min_{j \neq i} T_{ij} \sim \text{Exp}\left(\sum_{j \neq i} q_{ij}\right).$$

Hence the total rate of leaving i is

$$\boxed{\nu_i = \sum_{j \neq i} q_{ij} = -q_{ii}.}$$

Moreover,

$$\boxed{\mathbb{P}\left(\text{next state is } j \mid X(t) = i\right) = \frac{q_{ij}}{\sum_{k \neq i} q_{ik}} = \frac{q_{ij}}{\nu_i}.}$$

This reproduces the relation

$$q_{ij} = \nu_i p_{ij}.$$

4.8 Transition Semigroup and Matrix Exponential

For a finite-state continuous-time Markov chain, the transition matrix can be recovered from the generator.

Theorem 4.9. *Let Q be the generator of a finite-state continuous-time Markov chain. Then*

$$\boxed{P(t) = e^{tQ},}$$

where

$$e^{tQ} = \sum_{n=0}^{\infty} \frac{t^n Q^n}{n!}.$$

Thus the generator Q plays the role in continuous time that the transition matrix P plays in discrete time:

Discrete time	Continuous time
P^n	e^{tQ}.

4.9 Kolmogorov Backward Equations

The Chapman–Kolmogorov equations lead to differential equations for the transition probabilities.

Theorem 4.10 (Kolmogorov Backward Equations)**.** *The transition matrix satisfies*

$$\boxed{P'(t) = QP(t).}$$

Equivalently,

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t} p_{ij}(t) = \sum_{k \in S} q_{ik} p_{kj}(t).}$$

Proof. By the Chapman–Kolmogorov equations,

$$P(t+h) = P(h)P(t).$$

Hence

$$\frac{P(t+h) - P(t)}{h} = \frac{P(h) - I}{h} P(t).$$

Letting $h \downarrow 0$,

$$P'(t) = QP(t).$$

□

Remark 4.11. The backward equation acts through the *initial state* index:

$$\frac{\mathrm{d}}{\mathrm{d}t} p_{ij}(t) = \sum_{k} q_{ik} p_{kj}(t).$$

This explains the terminology "backward."

4.10 Kolmogorov Forward Equations

Theorem 4.12 (Kolmogorov Forward Equations)**.** *The transition matrix also satisfies*

$$\boxed{P'(t) = P(t)Q.}$$

Equivalently,

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}p_{ij}(t) = \sum_{k\in S} p_{ik}(t)q_{kj}.}$$

Proof. Using the Chapman–Kolmogorov equations in the form

$$P(t+h) = P(t)P(h),$$

we obtain

$$\frac{P(t+h) - P(t)}{h} = P(t)\frac{P(h) - I}{h}.$$

Letting $h \downarrow 0$,

$$P'(t) = P(t)Q.$$

□

Combining the two equations,

$$\boxed{P'(t) = QP(t) = P(t)Q.}$$

With the initial condition

$$P(0) = I,$$

the solution is

$$P(t) = e^{tQ}.$$

4.11 Evolution of an Initial Distribution

Suppose the initial distribution is the row vector

$$\alpha = (\alpha_i)_{i\in S}, \qquad \alpha_i = \mathbb{P}(X(0) = i).$$

The distribution at time t is

$$\boxed{\alpha(t) = \alpha P(t).}$$

Differentiating,

$$\alpha'(t) = \alpha P'(t) = \alpha P(t)Q,$$

and hence

$$\boxed{\alpha'(t) = \alpha(t)Q.}$$

These are the forward equations for the state probabilities.

4.12 Stationary Distributions

The notion of stationarity has a direct continuous-time analogue.

Definition 4.13 (Stationary Distribution)**.** A probability vector

$$\pi = (\pi_i)_{i \in S}$$

is called a *stationary distribution* if

$$\boxed{\pi P(t) = \pi}$$

for every $t \geq 0$.

For a finite-state continuous-time Markov chain, this condition can be expressed directly in terms of the generator.

Theorem 4.14. *A probability vector π is stationary if and only if*

$$\boxed{\pi Q = 0.}$$

Proof. If π is stationary, then

$$\pi P(t) = \pi.$$

Differentiating at $t = 0$,

$$\pi P'(0) = 0.$$

Since

$$P'(0) = Q,$$

we obtain

$$\pi Q = 0.$$

Conversely, suppose

$$\pi Q = 0.$$

Using

$$P(t) = e^{tQ},$$

we have

$$\pi P(t) = \pi \left(I + tQ + \frac{t^2 Q^2}{2!} + \cdots \right).$$

Since

$$\pi Q = 0,$$

we also have

$$\pi Q^n = 0 \qquad \text{for every } n \geq 1.$$

Therefore

$$\pi P(t) = \pi.$$

□

In coordinates, the stationary equations are

$$\boxed{\sum_{i \in S} \pi_i q_{ij} = 0, \qquad j \in S,}$$

together with

$$\boxed{\sum_{i \in S} \pi_i = 1.}$$

4.13 Global Balance Equations

Since

$$q_{jj} = -\sum_{k \neq j} q_{jk},$$

the equation

$$\sum_i \pi_i q_{ij} = 0$$

can be rewritten as

$$\boxed{\sum_{i \neq j} \pi_i q_{ij} = \pi_j \sum_{k \neq j} q_{jk}.}$$

The left-hand side represents the total equilibrium probability flow *into* state j, while the right-hand side represents the total equilibrium probability flow *out of* state j.

Thus stationarity corresponds to

$$\boxed{\text{total flow into each state} = \text{total flow out of each state.}}$$

4.14 Detailed Balance and Reversibility

A stronger condition is obtained by balancing the probability flow between each pair of states separately.

Definition 4.15 (Detailed Balance)**.** A probability distribution π satisfies the *detailed balance equations* if

$$\boxed{\pi_i q_{ij} = \pi_j q_{ji}}$$

for every $i, j \in S$.

Theorem 4.16. *If π satisfies the detailed balance equations, then π is a stationary distribution.*

Proof. For every j,

$$\begin{aligned}
\sum_{i \in S} \pi_i q_{ij} &= \sum_{i \neq j} \pi_i q_{ij} + \pi_j q_{jj} \\
&= \sum_{i \neq j} \pi_j q_{ji} - \pi_j \sum_{i \neq j} q_{ji} \\
&= 0.
\end{aligned}$$

Hence

$$\pi Q = 0,$$

so π is stationary. □

Remark 4.17. As in discrete time,

$$\boxed{\text{detailed balance} \Longrightarrow \text{stationarity},}$$

but the converse does not hold in general. Detailed balance is therefore a stronger condition than global balance.

4.15 Limiting Behavior

For an irreducible positive recurrent continuous-time Markov chain, the stationary distribution describes its long-run behavior under standard regularity assumptions.

Theorem 4.18 (Limiting Distribution)**.** *Suppose the continuous-time Markov chain is irreducible and positive recurrent, with stationary distribution π. Then*

$$\boxed{\lim_{t\to\infty} p_{ij}(t) = \pi_j}$$

for all $i, j \in S$, under the usual non-explosion assumptions.

Thus the limiting distribution becomes independent of the initial state.

Remark 4.19. Unlike discrete-time chains, continuous-time chains do not usually require a separate aperiodicity assumption. The random exponential holding times destroy the rigid integer-time oscillations responsible for periodicity in discrete time.

4.16 Birth–Death Processes

An important class of continuous-time Markov chains consists of processes that can move only between neighboring states.

Definition 4.20 (Birth–Death Process)**.** A *birth–death process* is a continuous-time Markov chain on

$$S = \{0, 1, 2, \ldots\}$$

whose only possible transitions are

$$i \longrightarrow i+1$$

and

$$i \longrightarrow i-1.$$

The transition rates are denoted by

$$q_{i,i+1} = \lambda_i, \qquad q_{i,i-1} = \mu_i,$$

where λ_i is the *birth rate* and μ_i is the *death rate*.

For $i \geq 1$,

$$q_{ii} = -(\lambda_i + \mu_i),$$

while at state 0,

$$q_{00} = -\lambda_0$$

when transitions to negative states are not allowed.

Thus the generator has the tridiagonal form

$$Q = \begin{pmatrix} -\lambda_0 & \lambda_0 & 0 & 0 & \cdots \\ \mu_1 & -(\lambda_1+\mu_1) & \lambda_1 & 0 & \cdots \\ 0 & \mu_2 & -(\lambda_2+\mu_2) & \lambda_2 & \cdots \\ \vdots & \vdots & \ddots & \ddots & \ddots \end{pmatrix}.$$

4.17 Stationary Distribution of a Birth–Death Process

For a birth–death process, the detailed balance equations take a particularly simple form:

$$\boxed{\pi_i \lambda_i = \pi_{i+1} \mu_{i+1}, \qquad i \geq 0.}$$

Hence

$$\pi_{i+1} = \pi_i \frac{\lambda_i}{\mu_{i+1}}.$$

Iterating,

$$\pi_1 = \pi_0 \frac{\lambda_0}{\mu_1},$$

$$\pi_2 = \pi_0 \frac{\lambda_0 \lambda_1}{\mu_1 \mu_2},$$

and in general

$$\boxed{\pi_n = \pi_0 \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n}, \qquad n \geq 1.}$$

The normalization condition

$$\sum_{n=0}^{\infty} \pi_n = 1$$

gives

$$\boxed{\pi_0 = \left[1 + \sum_{n=1}^{\infty} \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} \right]^{-1},}$$

provided that the series is finite.

Theorem 4.21. *A birth–death process has a stationary probability distribution of the above form if*

$$\boxed{\sum_{n=1}^{\infty} \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} < \infty.}$$

4.18 The Linear Birth–Death Process

A particularly important example is obtained when the birth and death rates are proportional to the current population size:

$$\lambda_n = n\lambda, \qquad \mu_n = n\mu.$$

Definition 4.22 (Linear Birth–Death Process)**.** A birth–death process satisfying

$$q_{n,n+1} = n\lambda, \qquad q_{n,n-1} = n\mu$$

is called a *linear birth–death process.*

If

$$X(t) = n,$$

then the total rate of the next event is

$$n(\lambda + \mu).$$

Hence the holding time in state n satisfies

$$\boxed{T_n \sim \mathrm{Exp}\Big(n(\lambda+\mu)\Big).}$$

Conditional on a jump occurring, the probability that it is a birth is

$$\boxed{\frac{\lambda}{\lambda+\mu},}$$

while the probability that it is a death is

$$\boxed{\frac{\mu}{\lambda+\mu}.}$$

Thus the embedded jump chain is a nearest-neighbor random walk whose upward and downward probabilities are

$$\frac{\lambda}{\lambda+\mu} \qquad \text{and} \qquad \frac{\mu}{\lambda+\mu}.$$

4.19 Pure Birth Process

If

$$\mu_n = 0$$

for every n, then only upward transitions are possible.

Definition 4.23 (Pure Birth Process)**.** A birth–death process satisfying

$$q_{n,n+1} = \lambda_n, \qquad q_{n,n-1} = 0$$

is called a *pure birth process*.

If the rates are constant,

$$\lambda_n = \lambda,$$

then the holding times are i.i.d.

$$\mathrm{Exp}(\lambda),$$

and the number of births by time t is a Poisson process.

If instead

$$\lambda_n = n\lambda,$$

the process is called a *Yule process*. The total birth rate increases linearly with the current population size.

4.20 Summary

A continuous-time Markov chain can be described by its holding rates

$$\nu_i$$

and jump probabilities

$$p_{ij},$$

or equivalently by its generator

$$Q = (q_{ij}),$$

where

$$\boxed{q_{ij} = \nu_i p_{ij}, \qquad i \neq j,}$$

and

$$\boxed{q_{ii} = -\nu_i.}$$

For small $h > 0$,

$$\boxed{P(h) = I + hQ + o(h).}$$

The transition matrices satisfy the semigroup property

$$\boxed{P(s+t) = P(s)P(t),}$$

and, for finite-state chains,

$$\boxed{P(t) = e^{tQ}.}$$

The Kolmogorov equations are

$$\boxed{P'(t) = QP(t)}$$

and

$$\boxed{P'(t) = P(t)Q.}$$

A stationary distribution satisfies

$$\boxed{\pi Q = 0, \qquad \sum_i \pi_i = 1.}$$

Detailed balance,

$$\boxed{\pi_i q_{ij} = \pi_j q_{ji},}$$

is a sufficient condition for stationarity.

For a birth–death process,

$$q_{i,i+1} = \lambda_i, \qquad q_{i,i-1} = \mu_i,$$

and detailed balance gives

$$\boxed{\pi_{i+1} = \pi_i \frac{\lambda_i}{\mu_{i+1}}.}$$

Thus continuous-time Markov chains retain much of the structure of discrete-time Markov chains, but the transition matrix is replaced by the generator Q, and the deterministic step number n is replaced by continuous time t.

Chapter 5

Brownian Motion and Its Basic Properties

5 Brownian Motion and Its Basic Properties

5.1 From Random Walks to Brownian Motion

Brownian motion can be viewed as the continuous scaling limit of a simple symmetric random walk.

Let

$$X_1, X_2, \dots$$

be independent and identically distributed random variables satisfying

$$\mathbb{P}(X_k = 1) = \mathbb{P}(X_k = -1) = \frac{1}{2}.$$

Define the simple symmetric random walk

$$S_0 = 0, \qquad S_n = \sum_{k=1}^{n} X_k.$$

For every fixed n,

$$\mathbb{E}[S_n] = 0, \qquad \mathrm{Var}(S_n) = n.$$

By the central limit theorem,

$$\boxed{\frac{S_n}{\sqrt{n}} \xrightarrow{\mathrm{d}} N(0,1).}$$

The central limit theorem describes the random walk at one fixed time. To obtain convergence of the entire path, we first embed the random walk into a space of continuous functions.

Define $S(u)$ for $u \geq 0$ by

$$S(k) = S_k, \qquad k = 0, 1, 2, \dots,$$

and interpolate linearly on every interval

$$[k, k+1].$$

Thus S is a random continuous function.

Theorem 5.1 (Donsker's Theorem)**.** *As $n \to \infty$,*

$$\boxed{\left\{\frac{1}{\sqrt{n}} S(nt) : t \geq 0\right\} \Longrightarrow \{B_t : t \geq 0\},}$$

where the convergence is weak convergence in the appropriate space of continuous functions, and $\{B_t : t \geq 0\}$ is a standard Brownian motion.

Remark 5.2. The central limit theorem gives convergence at a single time:

$$\frac{S_n}{\sqrt{n}} \xrightarrow{\mathrm{d}} N(0,1),$$

whereas Donsker's theorem gives convergence of the entire rescaled random path. It is therefore often regarded as a functional version of the central limit theorem.

5.2 Standard Brownian Motion

Definition 5.3 (Standard Brownian Motion)**.** A stochastic process

$$\{B_t : t \geq 0\}$$

is called a *standard Brownian motion* if:

1. $$B_0 = 0 \qquad \text{almost surely;}$$
2. it has independent increments: for every
$$0 \leq t_0 < t_1 < \cdots < t_n,$$
the random variables
$$B_{t_1} - B_{t_0}, \quad B_{t_2} - B_{t_1}, \quad \ldots, \quad B_{t_n} - B_{t_{n-1}}$$
are independent;
3. it has stationary Gaussian increments:
$$\boxed{B_{t+s} - B_s \sim N(0, t)}$$
for every $s, t \geq 0$;
4. the map
$$t \longmapsto B_t$$
is continuous almost surely.

Taking $s = 0$,

$$\boxed{B_t \sim N(0, t).}$$

Hence

$$\boxed{\mathbb{E}[B_t] = 0, \qquad \mathrm{Var}(B_t) = t.}$$

Remark 5.4. Brownian motion has both stationary and independent increments, just as the Poisson process does. Their path structures, however, are completely different: a Poisson process has jump paths, whereas Brownian motion has continuous paths.

5.3 Covariance Structure

The covariance structure of Brownian motion has a particularly simple form.

Proposition 5.5. *For every $s, t \geq 0$,*

$$\boxed{\mathbb{E}[B_s B_t] = s \wedge t = \min\{s, t\}.}$$

Proof. Assume without loss of generality that

$$0 \leq s \leq t.$$

Write

$$B_t = B_s + (B_t - B_s).$$

Then

$$\mathbb{E}[B_s B_t] = \mathbb{E}[B_s^2] + \mathbb{E}[B_s(B_t - B_s)].$$

Since $B_t - B_s$ is independent of B_s and has mean zero,

$$\mathbb{E}[B_s(B_t - B_s)] = \mathbb{E}[B_s]\,\mathbb{E}[B_t - B_s] = 0.$$

Therefore

$$\mathbb{E}[B_s B_t] = \mathbb{E}[B_s^2] = s.$$

Hence, for arbitrary $s, t \geq 0$,

$$\mathbb{E}[B_s B_t] = s \wedge t.$$

□

5.4 Gaussian Characterization

There is an equivalent way to characterize Brownian motion using Gaussian processes.

Definition 5.6 (Gaussian Process)**.** A stochastic process

$$\{X_t : t \in T\}$$

is called a *Gaussian process* if, for every

$$t_1, \ldots, t_n \in T,$$

the random vector

$$(X_{t_1}, \ldots, X_{t_n})$$

has a multivariate normal distribution.

Theorem 5.7 (Gaussian Characterization of Brownian Motion)**.** *A process*

$$\{B_t : t \geq 0\}$$

starting from $B_0 = 0$ is a standard Brownian motion if and only if:

1. *$\{B_t : t \geq 0\}$ is a Gaussian process;*
2. $$\mathbb{E}[B_t] = 0$$
   for every $t \geq 0$;
3. $$\boxed{\mathbb{E}[B_s B_t] = s \wedge t}$$
   for every $s, t \geq 0$;
4. *its sample paths are continuous almost surely.*

Proof. The forward implication follows from the defining properties of Brownian motion and the covariance calculation above.

Conversely, suppose the four stated properties hold. For $0 \leq s < t$, the increment

$$B_t - B_s$$

is Gaussian because it is a linear combination of jointly Gaussian random variables. Its mean is

$$\mathbb{E}[B_t - B_s] = 0,$$

and its variance is

$$\begin{aligned}\mathrm{Var}(B_t - B_s) &= \mathbb{E}[B_t^2] + \mathbb{E}[B_s^2] - 2\mathbb{E}[B_s B_t] \\ &= t + s - 2s \\ &= t - s.\end{aligned}$$

Hence

$$B_t - B_s \sim N(0, t-s).$$

It remains to verify independence of disjoint increments. Suppose

$$0 \le t_0 < t_1 \le t_2 < t_3.$$

Then

$$\begin{aligned}&\mathrm{Cov}\left(B_{t_1} - B_{t_0}, B_{t_3} - B_{t_2}\right) \\ &= \mathbb{E}[B_{t_1} B_{t_3}] - \mathbb{E}[B_{t_1} B_{t_2}] - \mathbb{E}[B_{t_0} B_{t_3}] + \mathbb{E}[B_{t_0} B_{t_2}] \\ &= t_1 - t_1 - t_0 + t_0 \\ &= 0.\end{aligned}$$

Since these increments are jointly Gaussian, zero covariance implies independence. The same argument applies to any finite collection of disjoint increments. Together with continuity, this gives all the defining properties of standard Brownian motion. □

Remark 5.8. For a general stochastic process, the mean and covariance do not determine its distribution. For a Gaussian process, however, all finite-dimensional distributions are completely determined by the mean and covariance functions.

Therefore a standard Brownian motion is characterized by

$$\boxed{m(t) = 0, \qquad K(s,t) = s \wedge t,}$$

together with continuity of its sample paths.

5.5 Transition Density

Suppose a Brownian motion is currently at position x. After an additional time t, its displacement is distributed as $N(0,t)$. Therefore the transition density is

Definition 5.9 (Brownian Transition Density)**.** For $t > 0$,

$$\boxed{p_t(x,y) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(y-x)^2}{2t}\right).}$$

Thus

$$B_t \mid \{B_0 = x\} \sim N(x,t),$$

and for $a < b$,

$$\boxed{\mathbb{P}_x(a < B_t < b) = \int_a^b p_t(x,y)\,\mathrm{d}y.}$$

More generally, by stationary increments,

$$B_t \mid \{B_s = x\} \sim N(x, t-s), \qquad 0 \le s < t.$$

5.6 Finite-Dimensional Distributions

Let

$$0 < t_1 < t_2 < \cdots < t_n.$$

The joint distribution of

$$(B_{t_1}, \ldots, B_{t_n})$$

can be written in terms of the transition densities.

Theorem 5.10 (Joint Density of Brownian Motion)**.** *For a standard Brownian motion starting from* 0*,*

$$\boxed{\begin{aligned} &f_{B_{t_1},\ldots,B_{t_n}}(y_1, \ldots, y_n) \\ &\qquad = p_{t_1}(0, y_1) p_{t_2-t_1}(y_1, y_2) \cdots p_{t_n-t_{n-1}}(y_{n-1}, y_n). \end{aligned}}$$

Proof. The increments

$$B_{t_1}, \quad B_{t_2} - B_{t_1}, \quad \ldots, \quad B_{t_n} - B_{t_{n-1}}$$

are independent, with respective distributions

$$N(0, t_1), \quad N(0, t_2 - t_1), \quad \ldots, \quad N(0, t_n - t_{n-1}).$$

Equivalently, conditional on $B_{t_{k-1}} = y_{k-1}$,

$$B_{t_k} \sim N(y_{k-1}, t_k - t_{k-1}).$$

Multiplying the corresponding conditional densities gives the stated formula. □

5.7 Path Regularity

Although Brownian paths are continuous, they are extremely irregular.

Theorem 5.11 (Irregularity of Brownian Paths)**.** *With probability one, a Brownian path*

$$t \longmapsto B_t$$

is continuous but is not monotone on any nontrivial interval and is nowhere differentiable.

Remark 5.12. Thus continuity should not be confused with smoothness. Brownian motion is continuous at every time but almost surely has no derivative at any time.

This extreme local irregularity is one of the reasons ordinary Riemann–Stieltjes calculus is insufficient for Brownian motion.

5.8 Variation of a Function

Let

$$\pi : \qquad s = t_0 < t_1 < \cdots < t_n = t$$

be a partition of $[s, t]$.

Definition 5.13 (p-Variation Along a Partition)**.** For $p > 0$, define

$$\boxed{V^{(p)}_{[s,t]}(f, \pi) = \sum_{i=0}^{n-1} |f(t_{i+1}) - f(t_i)|^p .}$$

The mesh size of the partition is

$$\boxed{\|\pi\| = \max_{0\le i\le n-1}(t_{i+1} - t_i).}$$

For $p = 1$,

$$V^{(1)}_{[s,t]}(f,\pi) = \sum_{i=0}^{n-1} |f(t_{i+1}) - f(t_i)|$$

is the variation of f along the partition.

For $p = 2$,

$$V^{(2)}_{[s,t]}(f,\pi) = \sum_{i=0}^{n-1} |f(t_{i+1}) - f(t_i)|^2$$

is the quadratic variation along the partition.

5.9 Quadratic Variation of Brownian Motion

The quadratic variation of Brownian motion is one of its most important pathwise properties.

Theorem 5.14 (Quadratic Variation)**.** *Let*

$$\{B_t : t \ge 0\}$$

be a standard Brownian motion, and let

$$\pi : \qquad s = t_0 < t_1 < \cdots < t_n = t$$

be a sequence of partitions satisfying

$$\|\pi\| \longrightarrow 0.$$

Then

$$\boxed{\sum_{i=0}^{n-1}(B_{t_{i+1}} - B_{t_i})^2 \longrightarrow t - s}$$

in probability.

Proof. Write

$$\Delta_i B = B_{t_{i+1}} - B_{t_i}, \qquad \Delta_i t = t_{i+1} - t_i.$$

Since Brownian increments are Gaussian,

$$\Delta_i B \sim N(0, \Delta_i t).$$

Hence

$$\mathbb{E}\Big[(\Delta_i B)^2\Big] = \Delta_i t.$$

Therefore

$$\mathbb{E}\left[\sum_i (\Delta_i B)^2\right] = \sum_i \Delta_i t = t - s.$$

For a centered Gaussian random variable $X \sim N(0, \sigma^2)$,

$$\mathbb{E}[X^4] = 3\sigma^4,$$

so

$$\mathrm{Var}(X^2) = 3\sigma^4 - \sigma^4 = 2\sigma^4.$$

Since the Brownian increments over disjoint intervals are independent,

$$\begin{aligned}\mathrm{Var}\left(\sum_i (\Delta_i B)^2\right) &= \sum_i \mathrm{Var}\left((\Delta_i B)^2\right) \\ &= 2\sum_i (\Delta_i t)^2 \\ &\le 2\|\pi\| \sum_i \Delta_i t \\ &= 2(t-s)\|\pi\|.\end{aligned}$$

As

$$\|\pi\| \longrightarrow 0,$$

the variance converges to zero. Hence

$$\sum_i (\Delta_i B)^2 \longrightarrow t - s$$

in L^2, and therefore also in probability. □

In particular, on $[0, t]$,

$$\boxed{[B]_t = t,}$$

where $[B]_t$ denotes the quadratic variation of Brownian motion.

5.10 Other Orders of Variation

The quadratic variation result is part of a more general pattern.

Theorem 5.15. *Let*

$$\pi : \qquad s = t_0 < t_1 < \cdots < t_n = t$$

with

$$\|\pi\| \longrightarrow 0.$$

For standard Brownian motion:

1. *if $p > 2$, then*

$$\boxed{V^{(p)}_{[s,t]}(B, \pi) \longrightarrow 0}$$

   in probability;
2. *if $p = 2$, then*

$$\boxed{V^{(2)}_{[s,t]}(B, \pi) \longrightarrow t - s}$$

   in probability;
3. *if $0 < p < 2$, then*

$$\boxed{V^{(p)}_{[s,t]}(B, \pi) \longrightarrow \infty}$$

   in probability.

Remark 5.16. The exponent $p = 2$ is therefore the critical variation exponent for Brownian motion:

$$\boxed{p > 2 : \ 0, \qquad p = 2 : \ t - s, \qquad 0 < p < 2 : \ \infty.}$$

In particular, taking $p = 1$, Brownian motion has infinite total variation on every nontrivial interval.

5.11 Brownian Scaling

Brownian motion is invariant under an appropriate rescaling of time and space.

Theorem 5.17 (Brownian Scaling)**.** *Let*

$$\{B_t : t \geq 0\}$$

be a standard Brownian motion. For every $c > 0$, define

$$\widetilde{B}_t = \frac{1}{\sqrt{c}} B_{ct}.$$

Then

$$\boxed{\{\widetilde{B}_t : t \geq 0\}}$$

is also a standard Brownian motion.

Proof. Clearly,

$$\widetilde{B}_0 = 0.$$

For $0 \leq s < t$,

$$\widetilde{B}_t - \widetilde{B}_s = \frac{1}{\sqrt{c}}(B_{ct} - B_{cs}).$$

Since

$$B_{ct} - B_{cs} \sim N(0, c(t-s)),$$

we obtain

$$\widetilde{B}_t - \widetilde{B}_s \sim N(0, t-s).$$

Independent increments are inherited from B, and continuity follows from the continuity of the original Brownian path. Therefore $\widetilde{B}$ is a standard Brownian motion. □

Equivalently,

$$\boxed{\{B_{ct} : t \geq 0\} \overset{\mathrm{d}}{=} \{\sqrt{c}\, B_t : t \geq 0\}.}$$

Remark 5.18. Brownian scaling expresses the characteristic relation

$$\boxed{\text{space scale} \sim \sqrt{\text{time scale}}.}$$

This is the continuous analogue of the $\sqrt{n}$-normalization appearing in the central limit theorem and Donsker's theorem.

5.12 Time Reversal on a Finite Interval

Brownian motion also possesses a time-reversal symmetry.

Theorem 5.19 (Time Reversal)**.** *Fix $T > 0$, and define*

$$\widetilde{B}_t = B_T - B_{T-t}, \qquad 0 \leq t \leq T.$$

Then

$$\boxed{\{\widetilde{B}_t : 0 \leq t \leq T\}}$$

is a standard Brownian motion on $[0, T]$.

Proof. We have

$$\widetilde{B}_0 = 0.$$

For

$$0 \le s < t \le T,$$
$$\widetilde{B}_t - \widetilde{B}_s = B_{T-s} - B_{T-t}.$$

This is a Brownian increment over an interval of length

$$(T-s) - (T-t) = t - s,$$

so

$$\widetilde{B}_t - \widetilde{B}_s \sim N(0, t-s).$$

Disjoint increments of $\widetilde{B}$ correspond to disjoint increments of B, and are therefore independent. Continuity is immediate from the continuity of B. Hence $\widetilde{B}$ is a standard Brownian motion. □

5.13 Time Inversion

Brownian motion has another useful symmetry involving inversion of time.

Theorem 5.20 (Time Inversion)**.** *Let*

$$\{B_t : t \ge 0\}$$

be a standard Brownian motion. Define

$$X_0 = 0, \qquad X_t = tB_{1/t}, \qquad t > 0.$$

Then

$$\boxed{\{X_t : t \ge 0\}}$$

is also a standard Brownian motion.

Proof. For $s, t > 0$,

$$\mathbb{E}[X_t] = 0,$$

and

$$\begin{aligned}\mathbb{E}[X_s X_t] &= st\,\mathbb{E}\left[B_{1/s}B_{1/t}\right] \\ &= st\left(\frac{1}{s} \wedge \frac{1}{t}\right) \\ &= s \wedge t.\end{aligned}$$

Since X is a Gaussian process, its finite-dimensional distributions agree with those of standard Brownian motion. The required continuity at $t = 0$ holds almost surely, so X is a standard Brownian motion. □

5.14 Law of the Iterated Logarithm

Brownian motion fluctuates on a scale slightly larger than $\sqrt{t}$ over long time intervals. The precise asymptotic envelope is described by the law of the iterated logarithm.

Theorem 5.21 (Law of the Iterated Logarithm)**.** *Let*

$$\{B_t : t \geq 0\}$$

be a standard Brownian motion. Then, almost surely,

$$\boxed{\limsup_{t\to\infty} \frac{B_t}{\sqrt{2t \log\log t}} = 1,}$$

and

$$\boxed{\liminf_{t\to\infty} \frac{B_t}{\sqrt{2t \log\log t}} = -1.}$$

Thus the asymptotic upper and lower envelopes of Brownian motion are approximately

$$\boxed{\pm\sqrt{2t \log\log t}.}$$

Remark 5.22. The law of the iterated logarithm refines the rough scale

$$|B_t| \sim \sqrt{t}.$$

It describes the largest fluctuations that occur infinitely often as $t \to \infty$, rather than the distribution of B_t at one fixed time.

5.15 Summary

Standard Brownian motion

$$\{B_t : t \geq 0\}$$

is characterized by

$$B_0 = 0,$$

stationary independent increments,

$$\boxed{B_t - B_s \sim N(0, t-s),}$$

and continuous sample paths.

Equivalently, Brownian motion is a continuous Gaussian process satisfying

$$\boxed{\mathbb{E}[B_t] = 0, \qquad \mathbb{E}[B_s B_t] = s \wedge t.}$$

Its transition density is

$$\boxed{p_t(x,y) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(y-x)^2}{2t}\right).}$$

Brownian paths are continuous but highly irregular. In particular, they are almost surely nowhere differentiable and have infinite total variation. Their quadratic variation, however, is finite and satisfies

$$\boxed{[B]_t = t.}$$

Brownian motion also possesses several fundamental symmetries:

$$\boxed{\frac{1}{\sqrt{c}} B_{ct} \stackrel{\mathrm{d}}{=} B_t,}$$

$$\boxed{B_T - B_{T-t} \stackrel{\mathrm{d}}{=} B_t,}$$

and

$$\boxed{t B_{1/t} \stackrel{\mathrm{d}}{=} B_t.}$$

Finally, its long-time fluctuations are described by

$$\boxed{\limsup_{t\to\infty} \frac{B_t}{\sqrt{2t \log\log t}} = 1}$$

almost surely, with the corresponding lower limit equal to -1.

Chapter 6

Martingales, Stopping Times, and Brownian Hitting Problems

6 Martingales, Stopping Times, and Brownian Hitting Problems

6.1 Filtrations

To describe information that becomes available over time, we use a filtration.

Definition 6.1 (Filtration)**.** A family of sigma-algebras

$$\{\mathcal{F}_t : t \geq 0\}$$

is called a *filtration* if

$$\mathcal{F}_s \subseteq \mathcal{F}_t \qquad \text{whenever } 0 \leq s \leq t.$$

The sigma-algebra $\mathcal{F}_t$ represents all information available up to time t.
For a stochastic process $\{X(t) : t \geq 0\}$, its natural filtration is

$$\boxed{\mathcal{F}_t^X = \sigma\{X(u) : 0 \leq u \leq t\}.}$$

Definition 6.2 (Adapted Process)**.** A stochastic process $\{X(t) : t \geq 0\}$ is said to be *adapted* to $\{\mathcal{F}_t\}$ if $X(t)$ is $\mathcal{F}_t$-measurable for every $t \geq 0$.

6.2 Martingales

A martingale models a process whose conditional expected future value, given all currently available information, is equal to its present value.

Definition 6.3 (Martingale)**.** A stochastic process

$$\{X(t) : t \geq 0\}$$

is called a *martingale* with respect to a filtration $\{\mathcal{F}_t\}$ if, for every $t, s \geq 0$,

1. $X(t)$ is $\mathcal{F}_t$-measurable;
2. $$\mathbb{E}[|X(t)|] < \infty;$$
3. $$\boxed{\mathbb{E}\left[X(t+s) \mid \mathcal{F}_t\right] = X(t)} \qquad \text{almost surely.}$$

Example 6.4 (Brownian Motion)**.** Standard Brownian motion

$$\{B_t : t \geq 0\}$$

is a martingale with respect to its natural filtration.

Indeed,

$$B_{t+s} = B_t + (B_{t+s} - B_t),$$

and $B_{t+s} - B_t$ is independent of $\mathcal{F}_t$ with mean zero. Hence

$$\begin{aligned}\mathbb{E}[B_{t+s} \mid \mathcal{F}_t] &= B_t + \mathbb{E}[B_{t+s} - B_t \mid \mathcal{F}_t] \\ &= B_t.\end{aligned}$$

Example 6.5. The process

$$\boxed{B_t^2 - t}$$

is a martingale.

Proof. For $s \geq 0$,

$$B_{t+s} = B_t + (B_{t+s} - B_t).$$

Hence

$$\begin{aligned}\mathbb{E}[B_{t+s}^2 \mid \mathcal{F}_t] &= B_t^2 + 2B_t\mathbb{E}[B_{t+s} - B_t \mid \mathcal{F}_t] \\ &\quad + \mathbb{E}[(B_{t+s} - B_t)^2 \mid \mathcal{F}_t] \\ &= B_t^2 + s.\end{aligned}$$

Therefore

$$\begin{aligned}\mathbb{E}[B_{t+s}^2 - (t+s) \mid \mathcal{F}_t] &= B_t^2 + s - t - s \\ &= B_t^2 - t.\end{aligned}$$

□

Example 6.6 (Exponential Martingale)**.** For every $\lambda \in \mathbb{R}$,

$$\boxed{\exp\left(\lambda B_t - \frac{\lambda^2 t}{2}\right)}$$

is a martingale.

Proof. Since

$$B_{t+s} = B_t + (B_{t+s} - B_t),$$

we have

$$\begin{aligned}&\mathbb{E}\left[e^{\lambda B_{t+s} - \lambda^2(t+s)/2} \mid \mathcal{F}_t\right] \\ &= e^{\lambda B_t - \lambda^2 t/2}\mathbb{E}\left[e^{\lambda(B_{t+s} - B_t) - \lambda^2 s/2}\right].\end{aligned}$$

Since

$$B_{t+s} - B_t \sim N(0, s),$$

its moment generating function gives

$$\mathbb{E}\left[e^{\lambda(B_{t+s} - B_t)}\right] = e^{\lambda^2 s/2}.$$

Hence

$$\mathbb{E}\left[e^{\lambda(B_{t+s} - B_t) - \lambda^2 s/2}\right] = 1,$$

and therefore

$$\mathbb{E}\left[e^{\lambda B_{t+s} - \lambda^2(t+s)/2} \mid \mathcal{F}_t\right] = e^{\lambda B_t - \lambda^2 t/2}.$$

□

6.3 The Markov Property

Definition 6.7 (Markov Process)**.** A stochastic process $\{X(t) : t \geq 0\}$ is called a *Markov process* if, for every suitable measurable function ϕ and every $s, t \geq 0$,

$$\boxed{\mathbb{E}\left[\phi(X(t+s)) \mid \mathcal{F}_t^X\right] = \mathbb{E}\left[\phi(X(t+s)) \mid X(t)\right]}$$

almost surely.

Thus, once the present state $X(t)$ is known, the entire past contains no additional information about the future.

Theorem 6.8. *Brownian motion is a Markov process.*

For a Brownian motion starting from x, let

$$p_t(x, y) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(y-x)^2}{2t}\right)$$

be its transition density. Then, for a suitable measurable function ϕ,

$$\boxed{\mathbb{E}\left[\phi(B_{t+s}) \mid \mathcal{F}_t\right] = \int_{\mathbb{R}} \phi(y) p_s(B_t, y)\, \mathrm{d}y.}$$

6.4 Stopping Times

A stopping time is a random time whose occurrence can be determined using only information available up to the present.

Definition 6.9 (Stopping Time)**.** A random variable

$$T : \Omega \to [0, \infty]$$

is called a *stopping time* with respect to $\{\mathcal{F}_t\}$ if

$$\boxed{\{T \leq t\} \in \mathcal{F}_t}$$

for every $t \geq 0$.

Informally, at time t one can determine whether the event

$$T \leq t$$

has already occurred without using any information from the future.

Example 6.10. For Brownian motion, define

$$T = \inf\{t \geq 0 : B_t = 1\}.$$

Then T is a stopping time because, by continuity,

$$\{T \leq t\} = \left\{\max_{0 \leq s \leq t} B_s \geq 1\right\},$$

which depends only on the Brownian path up to time t.

6.5 The Strong Markov Property

The ordinary Markov property applies at deterministic times. Brownian motion satisfies the stronger statement that the process can also be restarted at a stopping time.

Theorem 6.11 (Strong Markov Property)**.** *Let T be a stopping time for a standard Brownian motion $\{B_t : t \geq 0\}$. Define*

$$\widetilde{B}_t = B_{T+t} - B_T, \qquad t \geq 0.$$

Then

$$\boxed{\{\widetilde{B}_t : t \geq 0\}}$$

is a standard Brownian motion starting from 0, independent of the information available up to time T.

Remark 6.12. The strong Markov property means that, after a stopping time, Brownian motion probabilistically starts again from its current position and has no memory of the path before that time.

6.6 Hitting Times

One of the basic stopping times associated with Brownian motion is the first time at which the process reaches a specified level.

Definition 6.13 (Hitting Time)**.** For $x \in \mathbb{R}$, define

$$\boxed{T_x = \inf\{t > 0 : B_t = x\}.}$$

Proposition 6.14. *For every $x \in \mathbb{R}$, T_x is a stopping time.*

For $x \neq 0$, Brownian motion eventually reaches x almost surely:

$$\boxed{\mathbb{P}(T_x < \infty) = 1.}$$

6.7 The Reflection Principle

The distribution of T_x follows from the reflection principle.

Theorem 6.15 (Reflection Principle)**.** *Let*

$$M_t = \max_{0 \leq s \leq t} B_s.$$

For $x > 0$,

$$\boxed{\mathbb{P}(M_t \geq x) = 2\mathbb{P}(B_t \geq x).}$$

Equivalently,

$$\boxed{\mathbb{P}(T_x \leq t) = 2\mathbb{P}(B_t \geq x).}$$

Proof. Since

$$\{M_t \geq x\} = \{T_x \leq t\},$$

decompose according to the terminal position:

$$\mathbb{P}(T_x \leq t) = \mathbb{P}(T_x \leq t, B_t \geq x) + \mathbb{P}(T_x \leq t, B_t < x).$$

Reflect the part of the Brownian path after its first hitting time T_x across the horizontal line $y = x$. By the strong Markov property and symmetry of Brownian increments,

$$\mathbb{P}(T_x \le t, B_t < x) = \mathbb{P}(B_t > x).$$

Also,

$$\mathbb{P}(T_x \le t, B_t \ge x) = \mathbb{P}(B_t \ge x).$$

Hence

$$\mathbb{P}(T_x \le t) = 2\mathbb{P}(B_t \ge x).$$

□

6.8 Distribution of the Hitting Time

Theorem 6.16 (Hitting-Time Distribution)**.** *For $x \neq 0$ and $t > 0$,*

$$\boxed{\mathbb{P}(T_x \le t) = 2\left[1 - \Phi\left(\frac{|x|}{\sqrt{t}}\right)\right],}$$

where

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-u^2/2}\, \mathrm{d}u.$$

The density of T_x is

$$\boxed{f_{T_x}(t) = \frac{|x|}{\sqrt{2\pi t^3}} \exp\left(-\frac{x^2}{2t}\right), \qquad t > 0.}$$

Proof. Assume first that $x > 0$. By the reflection principle,

$$\mathbb{P}(T_x \le t) = 2\mathbb{P}(B_t \ge x).$$

Since

$$B_t \overset{\mathrm{d}}{=} \sqrt{t}\, Z, \qquad Z \sim N(0,1),$$

we obtain

$$\mathbb{P}(T_x \le t) = 2\mathbb{P}\left(Z \ge \frac{x}{\sqrt{t}}\right) = 2\left[1 - \Phi\left(\frac{x}{\sqrt{t}}\right)\right].$$

By symmetry, the same formula holds for $x < 0$ after replacing x by $|x|$. Differentiating with respect to t yields

$$f_{T_x}(t) = \frac{|x|}{\sqrt{2\pi t^3}} e^{-x^2/(2t)}.$$

□

Letting $t \to \infty$,

$$\mathbb{P}(T_x < \infty) = 1,$$

so one-dimensional Brownian motion hits every fixed point almost surely.

6.9 Mean and Laplace Transform of the Hitting Time

Although T_x is finite almost surely, its expectation is infinite.

Proposition 6.17. *For every $x \neq 0$,*

$$\boxed{\mathbb{E}[T_x] = \infty.}$$

Proof. Using the density,

$$\mathbb{E}[T_x] = \frac{|x|}{\sqrt{2\pi}} \int_0^\infty t^{-1/2} \exp\left(-\frac{x^2}{2t}\right) \mathrm{d}t.$$

As $t \to \infty$,

$$\exp\left(-\frac{x^2}{2t}\right) \longrightarrow 1,$$

so the integrand behaves like $t^{-1/2}$. Since

$$\int_1^\infty t^{-1/2} \, \mathrm{d}t = \infty,$$

the expectation diverges. □

Theorem 6.18 (Laplace Transform)**.** *For $x \neq 0$ and $\alpha > 0$,*

$$\boxed{\mathbb{E}\left[e^{-\alpha T_x}\right] = e^{-|x|\sqrt{2\alpha}}.}$$

Proof. Using the density of T_x,

$$\mathbb{E}\left[e^{-\alpha T_x}\right] = \frac{|x|}{\sqrt{2\pi}} \int_0^\infty t^{-3/2} \exp\left(-\alpha t - \frac{x^2}{2t}\right) \mathrm{d}t.$$

Evaluating this integral gives

$$\mathbb{E}\left[e^{-\alpha T_x}\right] = e^{-|x|\sqrt{2\alpha}}.$$

□

6.10 Brownian Motion Starting from an Arbitrary Point

Let

$$\{B_t^a : t \geq 0\}$$

denote Brownian motion starting from $a \in \mathbb{R}$:

$$B_t^a = a + B_t.$$

The hitting probability of any point x is one.
If $x \neq a$,

$$\boxed{\mathbb{P}_a(T_x < \infty) = \mathbb{P}_0(T_{x-a} < \infty) = 1.}$$

If $x = a$, then

$$\boxed{\mathbb{P}_a(T_a = 0) = 1}$$

when the hitting time is defined allowing time 0.

6.11 The Running Maximum

Definition 6.19 (Running Maximum)**.** Define

$$\boxed{M_t = \max_{0 \le s \le t} B_s.}$$

By the reflection principle, for $x > 0$,

$$\begin{aligned}\mathbb{P}(M_t \ge x) &= \mathbb{P}(T_x \le t) \\ &= 2\left[1 - \Phi\left(\frac{x}{\sqrt{t}}\right)\right].\end{aligned}$$

Theorem 6.20. *For every $t > 0$,*

$$\boxed{M_t \overset{\mathrm{d}}{=} |B_t|.}$$

Proof. For $x > 0$,

$$\mathbb{P}(M_t \ge x) = 2\left[1 - \Phi\left(\frac{x}{\sqrt{t}}\right)\right].$$

On the other hand, by symmetry of B_t,

$$\begin{aligned}\mathbb{P}(|B_t| \ge x) &= 2\mathbb{P}(B_t \ge x) \\ &= 2\left[1 - \Phi\left(\frac{x}{\sqrt{t}}\right)\right].\end{aligned}$$

Thus M_t and $|B_t|$ have the same distribution. □

The density of M_t is therefore

$$\boxed{f_{M_t}(x) = \sqrt{\frac{2}{\pi t}} \exp\left(-\frac{x^2}{2t}\right), \qquad x > 0.}$$

6.12 Joint Distribution of the Maximum and Terminal Value

The reflection principle gives more than the distribution of the maximum. It also gives the joint distribution of M_t and B_t.

Theorem 6.21. *For $x > 0$ and $y < x$,*

$$\boxed{\mathbb{P}(M_t \ge x, B_t \le y) = 1 - \Phi\left(\frac{2x - y}{\sqrt{t}}\right).}$$

Proof. On the event

$$\{T_x \le t\},$$

reflect the Brownian path after time T_x about the level x. A path satisfying

$$M_t \ge x, \qquad B_t \le y$$

is transformed into a Brownian path whose endpoint is at least

$$2x - y.$$

The reflection principle therefore gives

$$\mathbb{P}(M_t \geq x, B_t \leq y) = \mathbb{P}(B_t \geq 2x - y).$$

Since $B_t \sim N(0, t)$,

$$\mathbb{P}(B_t \geq 2x - y) = 1 - \Phi\left(\frac{2x - y}{\sqrt{t}}\right).$$

□

Differentiation gives the joint density.

Corollary 6.22. *For*

$$x > 0, \qquad y < x,$$

the joint density of (M_t, B_t) *is*

$$\boxed{f_{M_t, B_t}(x, y) = \frac{2(2x - y)}{\sqrt{2\pi t^3}} \exp\left(-\frac{(2x - y)^2}{2t}\right).}$$

6.13 Zeros of Brownian Motion

Definition 6.23 (Zero Set)**.** For a Brownian motion B, define

$$\boxed{\text{Zeros}(B) = \{t \geq 0 : B_t = 0\}.}$$

Let B^x denote Brownian motion starting from x.

Theorem 6.24. *For* $x \in \mathbb{R}$ *and* $t > 0$*,*

$$\boxed{\mathbb{P}\left(\text{Zeros}(B^x) \cap (0, t) \neq \varnothing\right) = 2\left[1 - \Phi\left(\frac{|x|}{\sqrt{t}}\right)\right].}$$

This follows immediately from the hitting-time formula, since reaching zero from x is equivalent to standard Brownian motion reaching $-x$.

6.14 Zeros in a Time Interval

For standard Brownian motion and

$$0 < a < b,$$

the probability that the process has at least one zero in the interval (a, b) is

$$\boxed{\mathbb{P}\left(\text{Zeros}(B) \cap (a, b) \neq \varnothing\right) = \frac{2}{\pi} \arccos\sqrt{\frac{a}{b}}.}$$

Equivalently,

$$\boxed{\mathbb{P}\left(\text{Zeros}(B) \cap (a, b) = \varnothing\right) = \frac{2}{\pi} \arcsin\sqrt{\frac{a}{b}}.}$$

Remark 6.25. These formulas show that the Brownian zero set is highly nontrivial. Brownian motion returns to zero repeatedly, even though the set of zero times has a complicated random structure.

6.15 The Last and Next Zero

Fix $t > 0$.

Definition 6.26. Define

$$\boxed{L_t = \sup\{0 \le s \le t : B_s = 0\}}$$

to be the last zero before time t, and define

$$\boxed{R_t = \inf\{s \ge t : B_s = 0\}}$$

to be the first zero after time t.

For $0 < x < t$,

$$\{L_t \le x\}$$

means that Brownian motion has no zero in the interval (x, t). Therefore

$$\boxed{\mathbb{P}(L_t \le x) = \frac{2}{\pi} \arcsin \sqrt{\frac{x}{t}}.}$$

For $y > t$,

$$\{R_t \ge y\}$$

means that Brownian motion has no zero in (t, y). Hence

$$\boxed{\mathbb{P}(R_t \ge y) = \frac{2}{\pi} \arcsin \sqrt{\frac{t}{y}}.}$$

More generally, for

$$0 < x < t < y,$$

the event

$$\{L_t \le x, R_t \ge y\}$$

means that Brownian motion has no zero throughout (x, y). Thus

$$\boxed{\mathbb{P}(L_t \le x, R_t \ge y) = \frac{2}{\pi} \arcsin \sqrt{\frac{x}{y}}.}$$

6.16 The Arcsine Law for the Last Zero

The distribution of L_t is one of the classical arcsine laws for Brownian motion.

Theorem 6.27 (Arcsine Law)**.** *For*

$$0 < x < t,$$

$$\boxed{\mathbb{P}(L_t \le x) = \frac{2}{\pi} \arcsin \sqrt{\frac{x}{t}}.}$$

Consequently, L_t has density

$$\boxed{f_{L_t}(x) = \frac{1}{\pi\sqrt{x(t-x)}}, \qquad 0 < x < t.}$$

Remark 6.28. The density becomes large near both 0 and t. Thus the last zero before time t is more likely to occur close to the beginning or close to the end of the interval than near its midpoint.

Taking $x \downarrow 0$,

$$\mathbb{P}(L_t \le 0) = 0,$$

and hence

$$\boxed{\mathbb{P}(L_t > 0) = 1.}$$

Therefore, almost surely, Brownian motion has zeros arbitrarily close to time 0.

6.17 Summary

A martingale satisfies

$$\boxed{\mathbb{E}[X(t+s) \mid \mathcal{F}_t] = X(t),}$$

and important Brownian martingales include

$$\boxed{B_t, \qquad B_t^2 - t, \qquad e^{\lambda B_t - \lambda^2 t/2}.}$$

A stopping time T is characterized by

$$\boxed{\{T \le t\} \in \mathcal{F}_t,}$$

and Brownian motion satisfies the strong Markov property:

$$\boxed{B_{T+t} - B_T}$$

is again a standard Brownian motion independent of the past.

For the hitting time

$$T_x = \inf\{t > 0 : B_t = x\},$$

the reflection principle gives

$$\boxed{\mathbb{P}(T_x \le t) = 2\left[1 - \Phi\left(\frac{|x|}{\sqrt{t}}\right)\right],}$$

and

$$\boxed{f_{T_x}(t) = \frac{|x|}{\sqrt{2\pi t^3}} e^{-x^2/(2t)}.}$$

Although

$$T_x < \infty \qquad \text{almost surely,}$$

we have

$$\boxed{\mathbb{E}[T_x] = \infty.}$$

The running maximum

$$M_t = \max_{0 \le s \le t} B_s$$

satisfies

$$\boxed{M_t \overset{\mathrm{d}}{=} |B_t|.}$$

Finally, the zero set of Brownian motion leads to the arcsine law

$$\boxed{\mathbb{P}(L_t \le x) = \frac{2}{\pi} \arcsin \sqrt{\frac{x}{t}}, \qquad 0 < x < t.}$$

Chapter 7

Brownian Motion with Drift and Brownian Bridges

7 Brownian Motion with Drift and Brownian Bridges

7.1 Brownian Motion with Drift

Standard Brownian motion has zero mean and variance t. A natural generalization allows both a deterministic drift and a different variance parameter.

Definition 7.1 (Brownian Motion with Drift)**.** A stochastic process

$$\{X(t) : t \geq 0\}$$

is called a *Brownian motion with drift coefficient* μ *and variance parameter* σ^2 if:

1. $$X(0) = 0;$$
2. $\{X(t) : t \geq 0\}$ has stationary and independent increments;
3. for every $t \geq 0$,
$$\boxed{X(t) \sim N(\mu t, \sigma^2 t).}$$
Thus, for $0 \leq s < t$,
$$\boxed{X(t) - X(s) \sim N\Big(\mu(t-s), \sigma^2(t-s)\Big).}$$

An equivalent representation follows immediately from standard Brownian motion.

Proposition 7.2. *Let*

$$\{B(t) : t \geq 0\}$$

be a standard Brownian motion. Then

$$\boxed{X(t) = \sigma B(t) + \mu t}$$

is a Brownian motion with drift coefficient μ *and variance parameter* σ^2*.*

Proof. Clearly,

$$X(0) = 0.$$

For $0 \leq s < t$,

$$X(t) - X(s) = \sigma(B(t) - B(s)) + \mu(t-s).$$

Since

$$B(t) - B(s) \sim N(0, t-s),$$

we obtain

$$X(t) - X(s) \sim N\Big(\mu(t-s), \sigma^2(t-s)\Big).$$

Stationarity and independence of increments follow directly from those of Brownian motion. □

In particular,

$$\boxed{\mathbb{E}[X(t)] = \mu t, \qquad \mathrm{Var}(X(t)) = \sigma^2 t.}$$

Remark 7.3. The term

$$\mu t$$

describes the deterministic trend, while

$$\sigma B(t)$$

describes the random fluctuation around that trend.

7.2 Conditional Distribution of Brownian Motion

We next study the distribution of Brownian motion at an intermediate time when its value at a later time is known.

Suppose

$$0 < s < t$$

and condition on

$$B_t = y.$$

Because (B_s, B_t) is jointly Gaussian, the conditional distribution of B_s given $B_t = y$ is again Gaussian.

Theorem 7.4 (Conditional Distribution)**.** *Let*

$$0 < s < t.$$

Given

$$B_t = y,$$

we have

$$\boxed{B_s \mid \{B_t = y\} \sim N\left(\frac{s}{t}y, \frac{s(t-s)}{t}\right).}$$

Proof. Since

$$\mathbb{E}[B_s] = \mathbb{E}[B_t] = 0,$$

and

$$\mathrm{Var}(B_s) = s, \qquad \mathrm{Var}(B_t) = t,$$

while

$$\mathrm{Cov}(B_s, B_t) = s,$$

the conditional mean of a jointly Gaussian pair is

$$\mathbb{E}[B_s \mid B_t = y] = \frac{\mathrm{Cov}(B_s, B_t)}{\mathrm{Var}(B_t)}y = \frac{s}{t}y.$$

The conditional variance is

$$\begin{aligned}\mathrm{Var}(B_s \mid B_t) &= \mathrm{Var}(B_s) - \frac{\mathrm{Cov}(B_s, B_t)^2}{\mathrm{Var}(B_t)} \\ &= s - \frac{s^2}{t} \\ &= \frac{s(t-s)}{t}.\end{aligned}$$

Hence

$$B_s \mid \{B_t = y\} \sim N\left(\frac{s}{t}y, \frac{s(t-s)}{t}\right).$$

□

The corresponding conditional density is

$$\boxed{f_{B_s|B_t}(x \mid y) = \frac{p_s(0,x)p_{t-s}(x,y)}{p_t(0,y)},}$$

where

$$p_t(x,y) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(y-x)^2}{2t}\right).$$

Substituting the Gaussian transition densities gives

$$\boxed{f_{B_s|B_t}(x \mid y) = \sqrt{\frac{t}{2\pi s(t-s)}} \exp\left[-\frac{t}{2s(t-s)}\left(x - \frac{s}{t}y\right)^2\right].}$$

Remark 7.5. The conditional mean

$$\frac{s}{t}y$$

is the point obtained by linearly interpolating between

$$(0,0) \qquad \text{and} \qquad (t,y).$$

Thus, after conditioning on the endpoint $B_t = y$, Brownian motion tends to fluctuate around the straight line joining the two prescribed endpoints.

7.3 Brownian Bridges

Conditioning Brownian motion to return to zero at a fixed terminal time leads to the Brownian bridge.

Definition 7.6 (Brownian Bridge)**.** A *Brownian bridge* on $[0,1]$ is the process

$$\boxed{\{Y(t) : 0 \le t \le 1\} \overset{\mathrm{d}}{=} \{B(t) : 0 \le t \le 1 \mid B(1) = 0\},}$$

where B is a standard Brownian motion starting from 0.

Thus a Brownian bridge begins at 0 and is constrained to return to 0 at time 1. From the conditional distribution formula, for $0 < t < 1$,

$$\boxed{Y(t) \sim N(0, t(1-t)).}$$

In particular,

$$\boxed{\mathbb{E}[Y(t)] = 0.}$$

7.4 Covariance of the Brownian Bridge

Since a Brownian bridge is a Gaussian process, its distribution is determined by its mean and covariance function.

Theorem 7.7. *Let*

$$\{Y(t) : 0 \le t \le 1\}$$

be a Brownian bridge. For

$$0 \le s \le t \le 1,$$

$$\boxed{\mathrm{Cov}(Y(s), Y(t)) = s(1-t).}$$

More generally,

$$\boxed{\mathrm{Cov}(Y(s), Y(t)) = (s \wedge t) - st.}$$

Thus

$$\mathrm{Var}(Y(t)) = t(1-t).$$

Notice that

$$\mathrm{Var}(Y(0)) = \mathrm{Var}(Y(1)) = 0,$$

which reflects the fact that both endpoints are fixed.

7.5 Construction from Standard Brownian Motion

A Brownian bridge can be constructed directly from an ordinary Brownian motion without explicitly conditioning on an event of probability zero.

Theorem 7.8 (Brownian Bridge Representation)**.** *Let*

$$\{B(t) : 0 \le t \le 1\}$$

be a standard Brownian motion and define

$$\boxed{B^*(t) = B(t) - tB(1), \qquad 0 \le t \le 1.}$$

Then

$$\{B^*(t) : 0 \le t \le 1\}$$

is a Brownian bridge.

Proof. Since $B^*(t)$ is a linear combination of jointly Gaussian random variables, $\{B^*(t)\}$ is a Gaussian process. Also,

$$\mathbb{E}[B^*(t)] = \mathbb{E}[B(t)] - t\mathbb{E}[B(1)] = 0.$$

For

$$0 \le s \le t \le 1,$$

we have

$$\begin{aligned}\mathbb{E}[B^*(s)B^*(t)] &= \mathbb{E}\left[(B(s) - sB(1))(B(t) - tB(1))\right] \\ &= \mathbb{E}[B(s)B(t)] - t\mathbb{E}[B(s)B(1)] \\ &\quad - s\mathbb{E}[B(t)B(1)] + st\mathbb{E}[B(1)^2].\end{aligned}$$

Since

$$\mathbb{E}[B(s)B(t)] = s,$$

$$\mathbb{E}[B(s)B(1)] = s,$$

$$\mathbb{E}[B(t)B(1)] = t,$$

and

$$\mathbb{E}[B(1)^2] = 1,$$

it follows that

$$\begin{aligned}\mathbb{E}[B^*(s)B^*(t)] &= s - st - st + st \\ &= s(1-t).\end{aligned}$$

Moreover,

$$B^*(0) = 0,$$

and

$$B^*(1) = B(1) - B(1) = 0.$$

Thus B^* is a centered Gaussian process with exactly the covariance function of a Brownian bridge. Hence

$$\{B^*(t) : 0 \le t \le 1\}$$

is a Brownian bridge. □

Remark 7.9. The expression

$$B(t) - tB(1)$$

removes from $B(t)$ the linear contribution associated with the random endpoint $B(1)$. The resulting process is forced to be zero at both $t = 0$ and $t = 1$.

7.6 Gaussian Processes and Uniqueness

The previous argument relies on an important fact about Gaussian processes.

Proposition 7.10. *The finite-dimensional distributions of a Gaussian process are uniquely determined by its mean function and covariance function.*

Thus, to identify two centered Gaussian processes, it is enough to verify that they have the same covariance function.

For the Brownian bridge,

$$\boxed{m(t) = 0,}$$

and

$$\boxed{K(s,t) = (s \wedge t) - st.}$$

7.7 Brownian Bridges on a General Time Interval

The same construction extends from $[0, 1]$ to $[0, T]$.

Let $T > 0$ and define

$$\boxed{B_T^*(t) = B(t) - \frac{t}{T}B(T), \qquad 0 \le t \le T.}$$

Then

$$B_T^*(0) = B_T^*(T) = 0,$$

and

$$\boxed{\mathrm{Cov}(B_T^*(s), B_T^*(t)) = (s \wedge t) - \frac{st}{T}.}$$

Therefore

$$\{B_T^*(t) : 0 \le t \le T\}$$

is a Brownian bridge from 0 to 0 over the interval $[0, T]$.

Remark 7.11. The original notes formulate the Brownian bridge on $[0, 1]$. The formula above is the same construction after a change of time scale.

7.8 Multidimensional Brownian Motion

Brownian motion can also be defined in higher-dimensional Euclidean spaces.

Definition 7.12 (Multidimensional Brownian Motion)**.** A d-dimensional Brownian motion is a process

$$\boxed{B(t) = \big(B^{(1)}(t), B^{(2)}(t), \ldots, B^{(d)}(t)\big),}$$

where

$$B^{(1)}, \ldots, B^{(d)}$$

are independent one-dimensional standard Brownian motions, started from their respective initial coordinates.

If the process starts from $x \in \mathbb{R}^d$, we write

$$\mathbb{P}_x$$

for its probability law.

Its increments satisfy

$$\boxed{B(t) - B(s) \sim N_d(0, (t-s)I_d),}$$

where I_d is the $d \times d$ identity matrix.

7.9 Hitting Points in Different Dimensions

The behavior of Brownian motion changes fundamentally with the dimension.

In one dimension, Brownian motion hits every fixed point almost surely:

$$\boxed{\mathbb{P}_0(T_x < \infty) = 1, \qquad x \in \mathbb{R}.}$$

In dimensions $d \geq 2$, a fixed nonzero point is not hit:

$$\boxed{\mathbb{P}_0(T_x < \infty) = 0, \qquad x \neq 0.}$$

This distinction motivates several notions of recurrence.

7.10 Point Recurrence and Neighborhood Recurrence

Definition 7.13 (Point Recurrence)**.** A process in $\mathbb{R}^d$ is called *point recurrent* if, almost surely, for every $x \in \mathbb{R}^d$, there exists a sequence

$$t_n \longrightarrow \infty$$

such that

$$\boxed{X(t_n) = x}$$

for all n.

Definition 7.14 (Neighborhood Recurrence)**.** A process in $\mathbb{R}^d$ is called *neighborhood recurrent* if, almost surely, for every

$$x \in \mathbb{R}^d$$

and every

$$\varepsilon > 0,$$

there exists a sequence

$$t_n \longrightarrow \infty$$

such that

$$\boxed{X(t_n) \in B(x, \varepsilon)}$$

for all n, where

$$B(x, \varepsilon) = \{y \in \mathbb{R}^d : \|y - x\| < \varepsilon\}.$$

Definition 7.15 (Transience)**.** A process is called *transient* if

$$\boxed{\|X(t)\| \longrightarrow \infty}$$

almost surely as $t \to \infty$.

7.11 Recurrence Classification by Dimension

Theorem 7.16 (Dimension Dependence of Brownian Motion)**.** *The recurrence behavior of d-dimensional Brownian motion is:*

1. *If*
$$d = 1,$$
then Brownian motion is point recurrent.
2. *If*
$$d = 2,$$
then Brownian motion is neighborhood recurrent but not point recurrent.
3. *If*
$$d \geq 3,$$
then Brownian motion is transient.

Thus

$d = 1$	:	point recurrent,
$d = 2$	:	neighborhood recurrent but not point recurrent,
$d \geq 3$	:	transient.

Remark 7.17. The two-dimensional case illustrates why point recurrence and neighborhood recurrence must be distinguished. Planar Brownian motion returns arbitrarily close to every point infinitely often, but the probability of hitting any particular prescribed point is zero.

7.12 Summary

A Brownian motion with drift coefficient μ and variance parameter σ^2 can be represented as

$$\boxed{X(t) = \sigma B(t) + \mu t.}$$

Hence

$$\boxed{X(t) \sim N(\mu t, \sigma^2 t).}$$

For standard Brownian motion and

$$0 < s < t,$$

conditioning on $B_t = y$ gives

$$\boxed{B_s \mid \{B_t = y\} \sim N\left(\frac{s}{t}y, \frac{s(t-s)}{t}\right).}$$

A Brownian bridge on $[0, 1]$ is Brownian motion conditioned on

$$B(1) = 0.$$

It is a centered Gaussian process with covariance

$$\boxed{\mathrm{Cov}(Y(s), Y(t)) = (s \wedge t) - st.}$$

An equivalent explicit construction is

$$\boxed{B^*(t) = B(t) - tB(1).}$$

Finally, the long-run behavior of Brownian motion depends critically on the spatial dimension:

$$\boxed{d = 1 : \text{ point recurrent,}}$$

$$\boxed{d = 2 : \text{ neighborhood recurrent but not point recurrent,}}$$

and

$$\boxed{d \geq 3 : \text{ transient.}}$$

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